



CRANFIELD UNIVERSITY

Master of Science Thesis

Computational Fluid Dynamics analysis of an aircraft antenna fairing

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Abstract

The increasing demand for in-flight connectivity has led to the integration of fuselage-mounted antennas on modern airliners, particularly for Ka-band satellite communication systems. While essential for passenger comfort and operational capability, such installations introduce aerodynamic penalties that may affect both performance and structural integrity.

This thesis presents a Computational Fluid Dynamics (CFD) study of a fuselage-mounted Ka-band antenna fairing. A simplified fuselage geometry was modeled in CATIA V5, and two-dimensional sectional analyses were performed in *ANSYS Fluent* at cruise conditions using a high-fidelity meshing strategy ($y^+ \approx 1$) and the SST $k-\omega$ turbulence model. The simulations focused on pressure coefficient distributions, velocity fields, and turbulent kinetic energy to assess aerodynamic loads and flow disturbances caused by the antenna installation.

The results indicate that the antenna fairing produces a localized adverse pressure gradient at the aft fillet, generating compact separation bubbles, wake regions, and increased turbulence intensity. These effects were strongest at forward sections and attenuated downstream, with flow recovery typically achieved within three to four antenna heights. The incremental drag was found to be dominated by compact form drag rather than extended wake effects.

The study demonstrates the effectiveness of a reduced-order 2D methodology for preliminary assessment of fuselage-mounted protuberances, balancing computational efficiency with physical fidelity. It also highlights the limitations of restricting the analysis to cruise conditions and simplified geometry. The findings provide valuable insight into antenna aerodynamics and establish a foundation for future work involving multi-condition simulations, three-dimensional modeling, and integration with structural analyses.

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Glossary

Term	Definition
CAD	Computer Aided Design
CFD	Computational Fluid Dynamics
RANS	Reynolds-Averaged Navier–Stoke
CPU	Central processing unit
STL	Stereo-Lithography
Cf	Friction coefficient
Cp	Pressure coefficient
aft	Rear

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1 Introduction

1.1 Context

The evolution of aviation in the 21st century mark down multiple criteria to make the sector more and more attractive.

The pandemic of COVID-19 put a significant stop to the activities but it restarted and still growing reaching a new high in 2024 of 4.8 billions individual passengers during the year. This is due by different aspect such as the safety in the sector. Indeed in 2023, it's the first year without any major incident in airliners causing fatalities.

But it's also due to the comfort of the aircraft. The comfort is one of the main criteria of passengers, with more and more technology on board, airlines need to maintain the improvement. For example the use Wi-Fi in-flight is more and more common in most of the national airlines, 62% of the global fleet have Wi-Fi services available during flight. This type of installation gives more and more challenge to engineers because of the complexity of these installation.

1.2 In-flight Wi-Fi

In-flight Wi-Fi is one of the major technological improvement of the 21st century, with the global increase of smartphones and laptops users, the need of Wi-Fi even during a flight came across relatively quickly. In fact, 97% of passengers use at least 1 electronic device during flight and Wi-Fi use can be mandatory for business travellers for example.

In-flight Wifi work by satellite, there is a big antenna on top of the aircraft that receives low orbit signal from Wi-Fi satellite constellation and redistribute the signal to the different passengers on the plane.

This antenna can vary in lot of factors, firstly in term of weight. This weight will vary in function of the type of the plane, larger plane will need a bigger field of use to give same stream to all the passengers and so the antenna will increase in scale. Also the speed of the stream will also change in function of the type of antenna, low orbit would usually be faster than high orbit but high orbit can give other informations.



Figure 1: Connectivity of an aircraft

1.3 Antenna assembly

Located at the top rear of the aircraft just before the aircraft vertical stabilizer, a specified zone that allow the antenna to have his best angle of reception without really modify the airworthiness of the plane. This installation can be really complex differentiating the type of antenna, the most common type is large antenna type that are assembled with a adaptater plate, this plate will do the link between aircraft skin and the antenna.

This adaptater plate is common to all installation but this adaptater plate could also be with a radome which is a structural kevlar cover that serve to protect aerodynamicly the antenna or potential bird strike. In our case the study will focus on small antenna installation, this kind of antenna only need adaptater plate because they are compact enough to need a aerodynamic fairing, they are also usually more aerodynamicly shaped than large antenna.

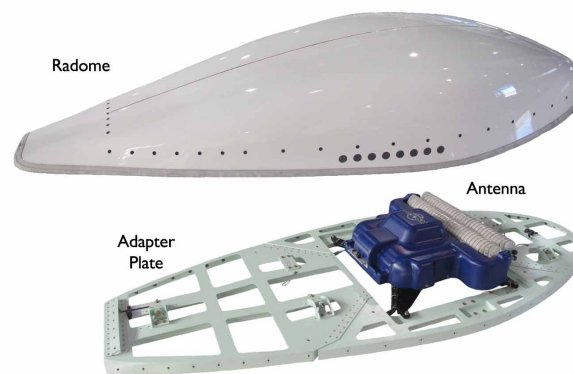


Figure 2: Example of antenna assembly

1.4 Objective of the study

The objective behind this study is to analyze the aerodynamic stream around the aircraft in order to find any pressure loads exerced on this antenna or any loads occurs on the aircraft structure that could differe from the original airplane and so impact the airwor-thiness of the aircraft.

In order to archieve this results we will use CATIA V5 for the design of the aircraft assembly with the antenna and in a second part we will use Ansys Fluent for the fluid dynamics simulation.

1.5 CATIA V5 & Ansys Fluent

CATIA V5 is a multi-platform software suite for computer-aided design (CAD), computer-aided manufacturing (CAM), computer-aided engineering (CAE), 3D modeling and prod-uct lifecycle management (PLM), developed by the French company Dassault Systèmes. CATIA benefits of a powerfull multi-users service that allow a great colaboration devel-opement from design to manufacturing. With multiple module of design (surface design, electronic design, solid design) it is one the of most multipurpose CAD software and it's used on most of the industrial sectors.

Ansys Fluent is one of the product of the american multinational Ansys Inc. Focused on Computational fluid dynamics analysis, Fluent is mostly used in R&D notably for its numbers of models used for CFD solving (two face flow, turbulence, combustion).

Usually Ansys Fluent work with other module because of its limit in terms of meshing, since the product doesn't get one, it is needed to couple it with the product Ansys Mechanical that integrate a mesher in its application.

1.6 Work organisation

This document will describe the study in different parts.

First part is focused on litterature around the subject and analysis of these differents document that will furnished our study.

Second part will be on the limits of the study, this have the objective to simplify study and make boundaries that restrict computer calculations.

In the part 3 will be the analysis and discussion around the project, that will figures all the case evaluated and the comparison between them.

And final part 4 that will cover the complete conclusion of the project and the scope of the future work needed to produce.

2 Literature review

2.1 Introduction

Fuselage-mounted antennas (such as blade antennas or dorsal satellite-communication radomes) are external protuberances that can significantly impact an aircraft's aerodynamics. These installations create additional parasite drag and can alter lift and moment characteristics, even if they are relatively small. For modern airliners cruising at transonic speeds (Mach 0.8 range), an improperly integrated antenna can produce flow disturbances (including shock waves and separation bubbles) that degrade performance [1, 2].

It is therefore crucial to analyze the aerodynamic loads (lift, drag, side-force) and pressure distribution on such antennas, as well as their influence on the aircraft's overall flow field and boundary layer behavior. This technical review summarizes the fundamental theories, historical milestones, and modern computational approaches for evaluating fuselage-mounted antenna aerodynamics, with emphasis on drag and lift characteristics, pressure cartography (pressure coefficient mapping), and boundary layer effects.

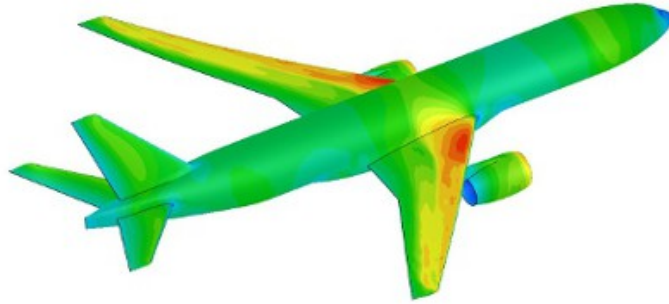


Figure 3: Aircraft CFD analysis

The discussion spans classical foundations from Prandtl's boundary layer theory to Eiffel's pioneering wind tunnel tests-up [3] to state-of-the-art CFD tools, turbulence modeling, and aerodynamic shape optimization techniques [4, 5]. Relevant studies and industry references are cited to support the analysis.

2.2 Fundamentals of Aerodynamic Loads and Boundary Layer Theory

Aerodynamic forces on any body arise from pressure and shear stress distributions over its surface. Lift is generated by pressure differences (lower pressure on the top of a body vs. bottom, for instance), while drag has two primary components: pressure drag due to flow separation and pressure imbalances, and skin-friction drag due to viscous shear in the boundary layer. When an antenna or radome is added to a fuselage, it contributes mostly to pressure drag (by presenting an additional bluff body to the flow) and a smaller increase in skin friction (due to added wetted area) [6].

The flow around the antenna-fuselage system is governed by the Navier-Stokes equations, which represent conservation of mass, momentum, and energy in a fluid. These equations generally have no closed-form solution for turbulent flows, so engineers rely on numerical methods (CFD) and turbulence modeling to solve them. In practice, the Reynolds-Averaged Navier-Stokes (RANS) equations with appropriate turbulence models are the workhorse for predicting aerodynamic loads on aircraft protuberances [7, 4]. RANS simulations introduce models (e.g. $k-\varepsilon$, $k-\omega$ SST, or Spalart–Allmaras) to approximate the effect of turbulent eddies, enabling efficient computation of wall-bounded flows at high Reynolds numbers typical of full-scale aircraft.

These models are well-established and robust for a wide range of flows, though their predictive accuracy for complex separated flows has improved only incrementally in recent decades. In most engineering applications, however, steady RANS with turbulence modeling strikes a good balance between accuracy and efficiency. Notably, regulatory guidance acknowledges the role of CFD: for example, EASA recommends CFD analysis to determine the pressure load distribution on antenna installations and to check effects on vibration and buffeting.

Understanding boundary layer behavior is fundamental to aerodynamic load analysis. Ludwig Prandtl’s boundary layer theory (1904) revealed that a thin layer of air near a surface transitions from zero velocity at the wall to free-stream velocity away from the wall, with viscosity playing a dominant role in this region [1]. Prandtl showed that treating flow as two regions an inner viscous boundary layer and an outer inviscid flow allowed engineers to finally predict drag and other aerodynamic effects that purely inviscid theory (e.g potential flow) could not explain.

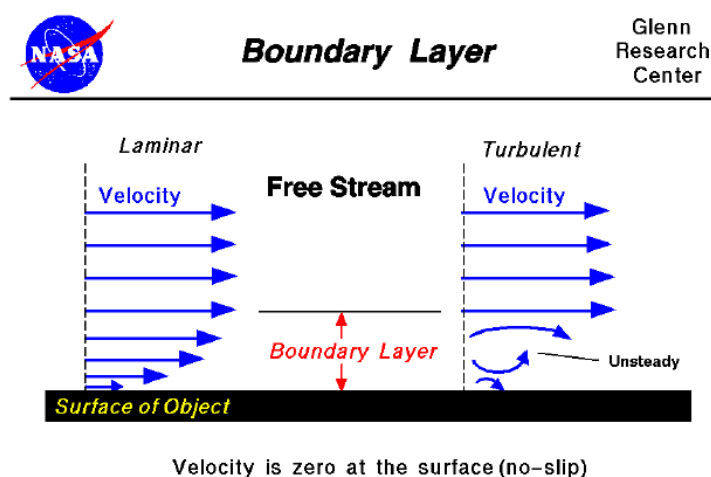


Figure 4: Boundary Layer

This was a breakthrough: it enabled quantitative calculation of skin-friction drag and explained how flow separation and pressure drag occur when the boundary layer slows down and detaches from the surface. Prandtl’s work directly informed the design of streamlined shapes to minimize drag for instance, demonstrating that a streamlined

fairing can delay flow separation, dramatically cutting pressure drag. These insights are directly relevant to antenna aerodynamics: a smoothly faired or low-profile antenna will tend to keep flow attached, whereas a bluff or abrupt antenna shape will induce a thicker wake and greater drag [6].

The boundary layer state (laminar vs. turbulent) also matters a laminar boundary layer has less skin friction but is more prone to separation, whereas a turbulent boundary layer has more friction but better energy mixing to resist separation. In practice, by the cruising stage of flight, the fuselage boundary layer is typically turbulent; any fuselage-mounted antenna will sit in this turbulent boundary layer or protrude through it, creating local disturbances. If the antenna is large relative to the boundary layer thickness, it can force the flow to separate and form a wake; if it's small or streamlined, it might only locally thicken the boundary layer. In either case, analyzing how the boundary layer adapts or separates is key to understanding the antenna's drag and any unsteady effects like buffeting.

2.3 Drag and Lift Characteristics of Fuselage-Mounted Antennas

Attaching an antenna or radome to an aircraft invariably incurs a drag penalty. The antenna disturbs the airflow on the fuselage: it increases the frontal area and causes additional pressure non-uniformity, resulting in extra drag. In aerodynamic terms, this is pure parasite drag (it does not contribute to lift, unlike a wing's induced drag). The drag increment depends on the antenna's size, shape, and placement [8].

A small blade antenna might cause a barely perceptible increase in cruise drag, whereas a large SATCOM radome hump can add a few percent to the aircraft's total drag in cruise. In one industry example, a streamlined Ku/Ka-band satcom radome (ThinKom's Ka1717) was shown via CFD to reduce drag by 75–90% compared to a flatter, blunt antenna enclosure; the streamlined shape effectively removed what would otherwise be the equivalent of up to 400 lb of parasitic weight in cruise due to drag [2].

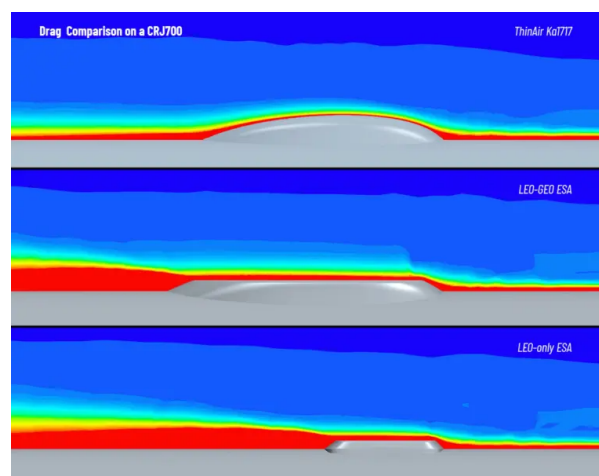


Figure 5: Antenna CFD comparison

This dramatic difference illustrates how sensitive drag is to antenna geometry: the flat-top design, despite a lower profile height, had sharp edges that provoked flow separation and even shock waves at cruise, leading to 5–10 $\times$ higher drag than the smoothly curved radome. In general, a well-contoured, aerodynamically smooth antenna fairing will incur much less drag than a bluff or boxy shape even if the latter has a slightly smaller frontal area. Designers often prefer conformal or low-profile antennas that blend into the fuselage surface to minimize such drag impact [8, 9].

Besides drag, a fuselage-mounted antenna can also affect the lift and stability of the aircraft, though usually to a minor degree. A symmetrically shaped antenna (like an ovoid radome) placed on top of the fuselage will not generate significant lift at zero angle of attack, but at higher aircraft angles of attack it can produce a small normal force [10]. Wind-tunnel tests on a large dorsal radome (ellipsoidal shape) showed that its contribution to lift was negligible at low angles of attack and only a slight increase at higher angles (as the radome presents a bigger projected area at high α).

However, a poorly shaped antenna could yield a much larger drag increment, as the ThinKom example (flat vs. curved radome) demonstrated. In summary, a fuselage-mounted antenna primarily contributes additional form drag, and secondarily some friction drag and potentially minor lift/side-forces. The exact aerodynamic coefficients depend on the shape and orientation. Designers strive to minimize the drag impact through careful shaping and placement. From a performance standpoint, even a 1% increase in total aircraft drag can cost millions in fuel over the life of an airliner, so antenna drag is by no means negligible. This has driven the development of aerodynamic fairings and low-drag radome designs.

On the flip side, structural engineers require accurate estimates of the aerodynamic loads on the antenna itself (forces and moments) to size the antenna's mounting and the fuselage reinforcements. At critical design conditions (e.g. dive speed, high angle of attack, gusts), the antenna will experience significant airloads. For instance, on a large surveillance radome, knowing the pressure distribution across its surface is essential for its structural design. We turn next to how these pressure distributions (pressure cartography) are analyzed.

2.4 Pressure Cartography and Boundary Layer Behavior Around Antennas

Fuselage-mounted Ka-band SATCOM radomes that are directly attached to the airframe alter the local airflow, creating distinctive pressure coefficient distributions over the radome and nearby fuselage. CFD simulations and wind-tunnel tests show that a radome acts as a small hump on the fuselage: flow stagnates at the front of the radome (high pressure coefficient near +1.0), then accelerates over the top, causing a suction peak (low pressure) on the forward upper surface, before decelerating and recovering pressure towards the aft end [8, 9].

Crucially, the presence of a radome can thicken and disturb the aircraft's boundary layer. If the radome has a bluff or steep aft shape, flow separation may occur just behind it, creating a wake region of recirculating flow and increased drag. CFD studies emphasize

using turbulence models capable of capturing this: for instance, the realizable k - model was chosen in one radome optimization study specifically because it can handle “boundary layer separations that may occur behind a bluff-like structure such as our radome,” whereas a standard k - model would have been more computationally expensive [7, 4].

This indicates that flow separation behind the radome (and the resulting thicker wake) is a real concern if the radome is not aerodynamically well-shaped. A longer, more streamlined radome shape can prevent flow separation by allowing a gentler pressure recovery. In fact, design guidelines from earlier research show that making the radome as slender as possible greatly mitigates adverse pressure gradients: a 1990 study by Girgosian et al. found that a higher length-to-height (L/H) ratio and a smooth tail angle result in substantially lower drag, as they reduce flow separation and pressure drag [9]. Other shape factors like the radome’s wetted area and width also correlate with its pressure footprint and drag impact.

Because these radomes sit on the aircraft exterior, they also interact with the fuselage’s pre-existing boundary layer. For large airliners at cruise, the fuselage boundary layer can be several centimeters thick on the upper fuselage where antennas are mounted. A flush or low-profile radome (only a few inches tall) might be partly submerged in this boundary layer. In such cases, much of the radome is in slower-moving air, which can reduce peak dynamic pressure on it but if it protrudes beyond the boundary layer, it effectively presents a bluff obstacle to the free stream. Modern design trends therefore aim to keep these SATCOM antenna structures as low-profile as possible to minimize protrusion. The European ACASIAS research project explicitly targeted conformal (“flush”) antennas integrated into the fuselage skin to eliminate large external humps[11].

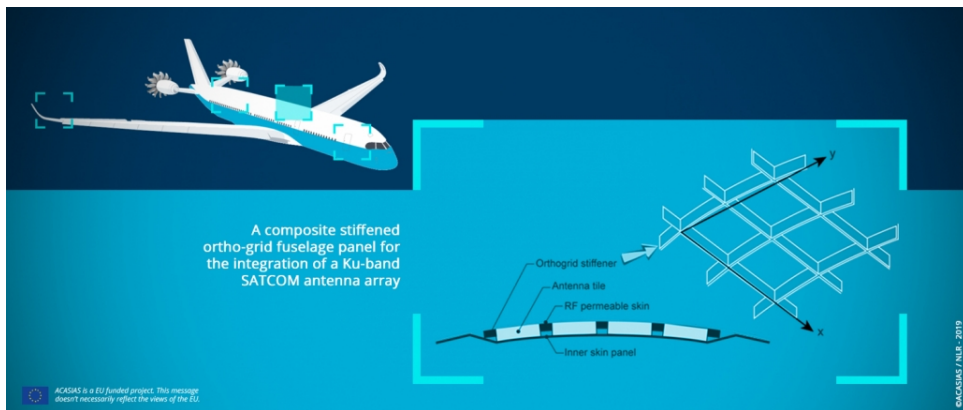


Figure 6: ACASIAS System

By embedding the antenna within a smooth composite fuselage panel (with a locally RF-transparent skin), the outer mold line remains clean. This approach avoids the pronounced pressure disturbances that a traditional radome hump would create, and thus yields nearly undisturbed flow over the fuselage. In essence, replacing a protruding radome with a flush integration can almost eliminate the pressure drag penalty of the antenna: for example, CFD comparisons of an airliner with and without a low-profile radome showed a measurable drag penalty for the external radome, which translates to extra fuel burn and emissions, whereas a conformal antenna incurs virtually no drag increase.

2.5 Reynolds Number and Boundary Layer Characteristics

In external aerodynamics, the Reynolds number (Re) sets the scale of the flow and the nature of the boundary layer around the aircraft fuselage and any mounted antenna. It is defined as $Re = \frac{\rho U L}{\mu} = \frac{U L}{\nu}$, where U is a characteristic flow velocity (e.g. cruise speed), L a characteristic length (e.g. fuselage length or antenna diameter), and ν the kinematic viscosity. For a typical airliner in cruise, Re based on fuselage length (say $L \sim 30$ m) can be on the order of 10^7 – 10^8 , indicating a fully turbulent flow over most of the fuselage and any protruding antennas [6]. Even a small antenna (length scale $L \sim 0.1$ – 1 m) at cruise can have Re in the millions, so its flow is also likely turbulent.

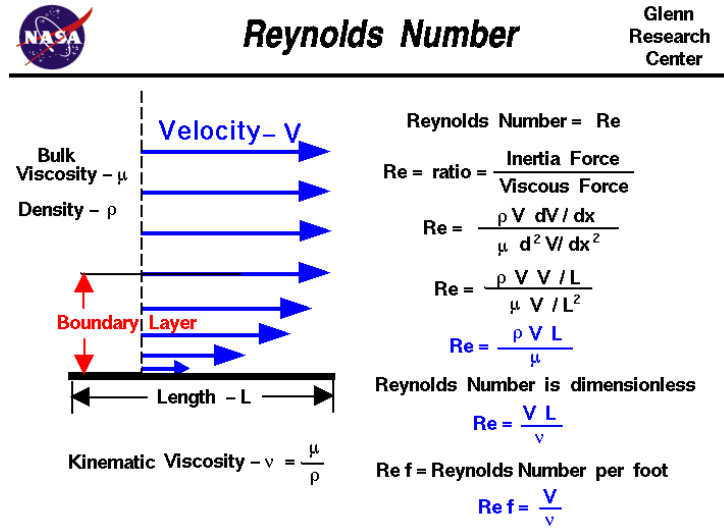


Figure 7: Reynolds calculations

High Reynolds numbers imply relatively thin boundary layers (since viscous effects are confined to a thin region near the surface) and high velocity gradients at the wall. Thus, resolving the boundary layer around the fuselage and antenna is critical. The boundary layer thickness δ grows with distance along the fuselage, but at high Re it remains small relative to body size (e.g. on the order of centimeters over a meter-sized antenna).

As a first estimate, empirical flat-plate correlations (e.g. $\delta \approx 0.37 L Re_L^{-1/5}$ for a turbulent boundary layer) can be used to predict boundary layer thickness on the fuselage or antenna surface. This estimate helps determine how far outward the mesh must resolve flow to capture velocity gradients. Notably, because each aircraft component has its own relevant length scale and local Re , mesh sizing should be determined for each component individually. For example, the fuselage Re may dictate a different first-cell height than the antenna Re , since a small antenna in the same freestream has a lower Reynolds number and a relatively thicker viscous sublayer in absolute terms. It is crucial to account for these differences to accurately capture the boundary layer on all surfaces.

High Reynolds number flows around antennas ensure the boundary layer is turbulent (unless a laminar flow control is in effect, which is uncommon on such small appendages). A turbulent boundary layer has higher skin friction and fuller velocity profiles than a laminar one, influencing antenna drag and flow separation. The high Re also means strong inertial forces dominate over viscosity in the outer flow, so flow features like the

antenna's wake can persist far downstream. However, near the wall the viscosity-affected layer (boundary layer) still needs careful resolution. The importance of Re manifests in how fine the mesh must be near walls: at higher Re , the viscous sublayer is thinner and wall shear stress higher, requiring smaller grid spacing (in wall units) to resolve the same physics [12].

Before meshing, one typically estimates the required first cell height Δy_1 for a desired wall resolution using flat-plate theory: for example, assuming a friction coefficient C_f (from an empirical formula like $C_f \approx 0.026 Re_L^{-1/7}$ for turbulent flow) to compute the wall shear stress $\tau_w = \frac{1}{2}\rho U^2 C_f$, then the friction velocity $u_\tau = \sqrt{\tau_w/\rho}$. The cell height for a target y^+ is given by $\Delta y_1 = y^+ \frac{\nu}{u_\tau}$. By performing this calculation for the fuselage surface and for the antenna surface (using their respective length-based Re), the mesh can be designed to meet y^+ criteria on both. It is advised to also estimate the total boundary layer thickness δ (e.g. from flat plate correlation) on those surfaces and ensure the prism layer mesh extends beyond δ with sufficient layers. A safety factor (e.g. $1.5\times$ the flat-plate δ) is often applied to account for 3D effects. This guarantees the mesh fully captures the boundary layer and any velocity gradients or transition effects around the antenna and fuselage.

Finally, the Reynolds number sensitivity ties into flow regime and separation: at very high Re , the tendency for flow separation (e.g. behind the antenna) can increase due to thinner, more turbulent boundary layers being susceptible to adverse pressure gradients. Accurately predicting separation locations (like where the antenna wake or a shock-induced separation might occur) demands fine resolution. In summary, knowing the Re and resulting boundary layer properties around a fuselage-mounted antenna guides the mesh spacing and domain extent required: a high- Re turbulent flow demands a fine near-wall mesh and adequate downstream refinement to capture the thin boundary layer and long wake generated by the antenna.

2.6 y^+ and Near-Wall Mesh Resolution for RANS

A crucial parameter for near-wall meshing is the non-dimensional wall distance, y^+ , which indicates how coarse or fine the mesh is relative to the viscous length scale. It is defined as $y^+ = \frac{u_\tau y}{\nu}$, where y is the distance from the wall (to the cell center), and $u_\tau = \sqrt{\tau_w/\rho}$ is the friction velocity related to wall shear stress. Essentially, y^+ can be interpreted as a local Reynolds number for the distance y from the wall. The magnitude of y^+ determines the relative importance of viscous vs. turbulent momentum transfer in the cell next to the wall.

In turbulent boundary layers, the near-wall region is classically divided into three layers: the viscous sublayer (where laminar viscous shear dominates, roughly $y^+ < 5$), the buffer layer ($5 < y^+ < 30$, a blending region of both viscous and turbulent stresses), and the log-law layer (fully turbulence-dominated, $y^+ > 30$ up to a few hundred). This is illustrated by the law-of-the-wall: in the sublayer, velocity increases linearly with y^+ ($u^+ \approx y^+$), whereas in the log-law region $u^+ = \frac{1}{\kappa} \ln(y^+) + C$ (with $\kappa \approx 0.41$, $C \approx 5$).

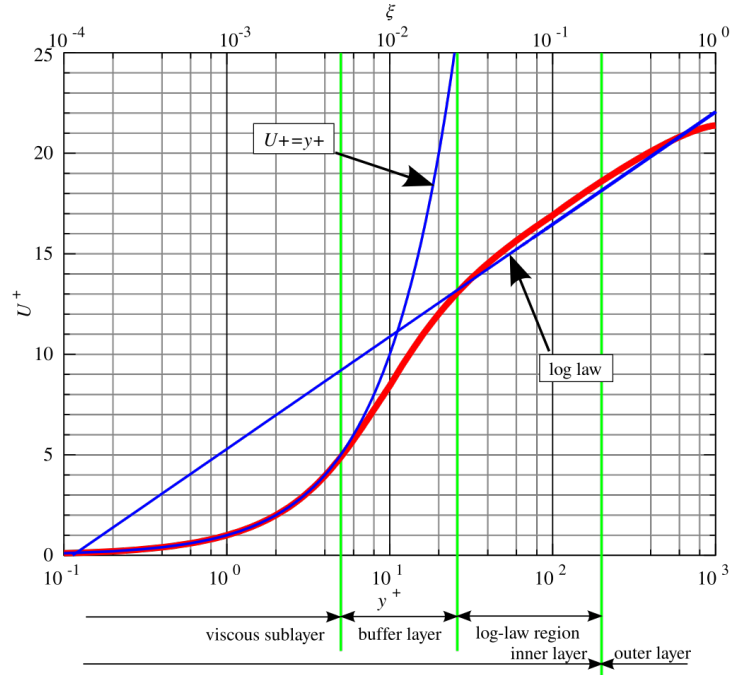


Figure 8: Law of Wall

Mesh resolution is judged by where the first cell center lies in this y^+ spectrum, and turbulence models have specific requirements for optimal y^+ placement. Reynolds-Averaged Navier–Stokes (RANS) simulations typically use one of two wall-treatment approaches, often referred to as “low- y^+ ” (resolving the viscous sublayer) or “high- y^+ ” (using wall functions). In a low- y^+ approach, the mesh resolves the entire boundary layer down to the viscous sublayer; this means the first cell center should ideally be at $y^+ \approx 1$ (on the order of unity).

Most modern aerodynamic RANS models (e.g. the $k-\omega$ SST model or Spalart–Allmaras) are formulated as low-Reynolds-number models that require $y^+ 1\text{--}2$ for full accuracy [4, 5]. Menter’s SST, for instance, recommends $y^+ < 3$ for the first cell, and studies have found that values below 2 are needed to match experimental skin friction and flow separation data. When the viscous sublayer is resolved, one must also use a sufficiently fine gradation of layers (typically 20–40 prism layers with growth ratio 1.1) to cover the boundary layer thickness[7].

This fully resolves the wall shear stress profile and generally yields more accurate predictions of wall friction, pressure gradients, and flow separation in adverse pressure gradient flows. RANS simulations of aircraft often favor the low- y^+ approach because it improves the fidelity of boundary layer calculations important for drag and maximum lift prediction. Notably, if flow separation is expected (as behind an antenna), resolving $y^+ \approx 1$ is advised to correctly capture the onset of separation.

By contrast, the high- y^+ approach relies on wall functions to bridge the viscous sublayer. Here the first cell is placed in the log-law region, typically $30 < y^+ < 300$. The idea is to avoid the buffer layer; a common guideline is to aim for $y^+ \approx 30$ or higher so that the cell is fully in the logarithmic region.

For example, standard $k-\epsilon$ models often use wall function formulas that assume the first node is at $y^+ \sim 50$ (in the log layer). If y^+ is too low (say 5–10) while using a wall function, the solver’s assumptions break down essentially you’d be using a wall function in a region where viscosity is still important.

Likewise too high a y^+ (greater than a few hundred) means the first cell is so far out that it misses a significant part of the inner boundary layer (reducing accuracy). Thus, a valid wall-function mesh will have y^+ in an intermediate range (30 to 300 is often quoted). In practice, for very high Reynolds numbers (e.g. $> 10^7$), even wall-function meshes may target the lower end (30–100) because extremely large y^+ can degrade accuracy.

A major advantage of the high- y^+ approach is that it needs far fewer cells near the wall (since the viscous sublayer isn’t resolved), allowing larger time-steps or less cells in steady simulations. However, it may struggle to predict separation or surface pressures in complex flows since the modeling of the near-wall region is empirical. In the context of an aircraft antenna, using wall functions might be acceptable for attached flow, but if the antenna induces flow separation or significant curvature effects, a low- y^+ mesh is generally more reliable.

Comparison of near-wall mesh strategies. Top: Low- y^+ approach the first cell lies inside the viscous sublayer (red shaded region), requiring many fine prism layers to resolve the velocity gradient to the wall. Bottom: High- y^+ approach the mesh uses a few thicker layers and places the first cell in the logarithmic region (red), using a wall function to model the inner layer. In general, low- y^+ meshes yield higher fidelity in predicting wall shear and separation, at the cost of more cells, whereas high- y^+ meshes are coarser but rely on turbulence model approximations for the wall region.

3 Conception and modeling

3.1 Computational Time and Meshing Strategy in Antenna Aerodynamics

Computational Fluid Dynamics (CFD) simulations of fuselage-mounted antennas can require significant computational resources due to the fine mesh resolution needed to accurately capture complex flow phenomena such as boundary layer development, flow separation, and shock formation in transonic regimes. The total computational time depends on several factors:

- **Mesh size and topology:** A finer mesh, particularly in regions of high curvature or near walls, increases the number of cells and therefore the total number of equations to be solved.
- **Turbulence modeling:** Higher-fidelity turbulence models generally require more iterations per time step and additional transport equations.
- **Solver settings:** Smaller time steps or tighter convergence criteria improve accuracy but increase runtime.
- **Hardware performance:** The number of available CPU cores, clock speed, and memory bandwidth directly impact simulation turnaround time.

The computational cost T_{comp} can be expressed in simplified form as:

$$T_{comp} \propto N_{cells} \times N_{iterations} \times C_{per\ cell}$$

where N_{cells} is the total number of mesh cells, $N_{iterations}$ is the number of solver iterations to convergence, and $C_{per\ cell}$ is the average computational cost per cell per iteration. For steady-state RANS solutions, reducing N_{cells} while preserving mesh quality in critical regions is the most direct way to decrease computational time.

Meshing Strategy with $y^+ \approx 1$ and 2D Reduction:

In this study, the near-wall mesh is designed to achieve a target $y^+ \approx 1$, ensuring that the first cell height lies within the viscous sublayer of the turbulent boundary layer. This high-resolution approach avoids reliance on wall functions and enables direct integration of the wall shear stress for accurate prediction of skin friction drag and boundary layer development around the antenna.

While such a mesh strategy greatly improves fidelity in the near-wall region, it can also lead to a significant increase in the total number of cells in a full three-dimensional simulation. To manage computational cost, the aerodynamic cartography was reduced to a two-dimensional (2D) slice analysis. This 2D approach focuses on a longitudinal plane intersecting the antenna's centerline, allowing detailed mapping of the surface pressure coefficient C_p and boundary layer profiles without resolving the entire 3D fuselage flow field.

The reduction from 3D to 2D decreases mesh cell count by orders of magnitude, substantially lowering solver memory requirements and wall-clock time, while preserving accuracy

for sectional load analysis and pressure distribution studies in symmetric flow conditions. This compromise is particularly effective when investigating laminar-to-turbulent transition locations, local separation, and stagnation point movement on radomes or fuselage-mounted antennas, provided that spanwise flow effects and crossflow instabilities are negligible in the chosen configuration. **Additional Runtime Optimization:** Beyond

mesh reduction, computational time can be further minimized by:

- Using structured or hybrid grids to improve solver convergence.
- Applying local mesh refinement only where gradients are steep (adaptive meshing).
- Employing parallel processing across multiple CPU cores.
- Using convergence acceleration techniques such as multigrid solvers.

These strategies allow for maintaining high accuracy in regions of interest while significantly reducing overall computational cost.

3.2 CAD Modeling

We start the work by modeling the assembly between the aircraft and the antenna.

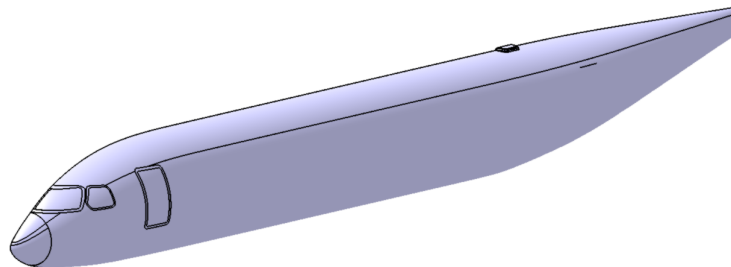


Figure 9: CAD Assembly

The work on the aircraft model was clear, try to simplify as much as possible the model in order to have less of part to merge. The model is also split in half due to aircraft symmetry. Finally the study will focus on the fuselage geometry only, wing and stabilizer turbulence will not be include in the study.

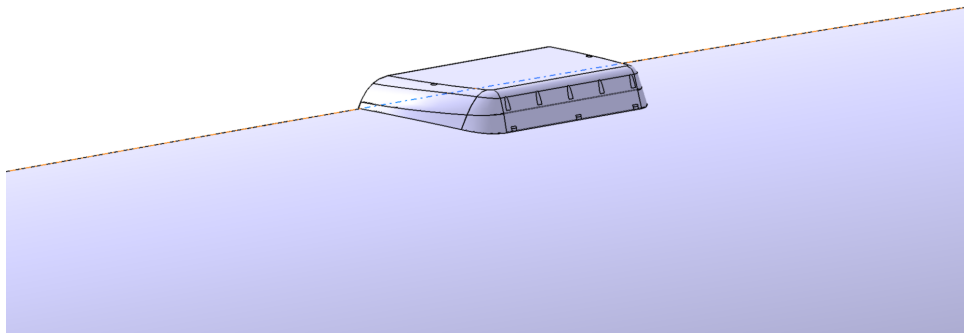


Figure 10: Antenna close-up

Here is a close view of the antenna, all fixing point have been removed in order to smoother the model. Gaps and holes can create turbulence and impact baddly the simulation, this is way we always search for the simplest model.

Now that we have our 3D model, the strategy is to cut in couple 2D sections the 3D model to only work with 2D Sections in the simulation. We will create a section 0 (half the aircraft), section 50mm, section 100mm, 150, 200 and 250mm. Theses sections will help you to create a good cartography of the boundary layer and all the pressures exerced on the aircraft antenna.

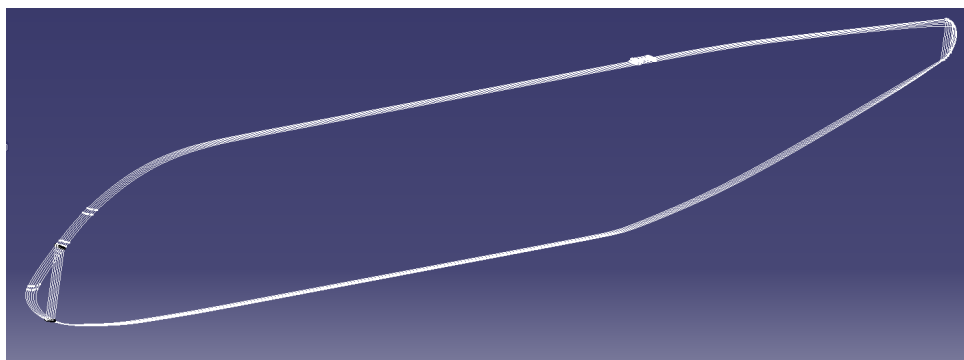


Figure 11: Intersection

So we design the outline of the sections, one for every plane by using the intersection function of CATIA Generative Shape Design module. From these intersections we create sketch that match intersections. Due to the imperfect surface, some gaps on the geometry need to be fixed.

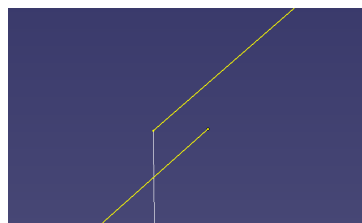


Figure 12: Gaps

After creating all the sketches, we can create the surface using Fill function and we obtain the section wanted. We will create sections with and without antenna to create a comparison between the two simulation.

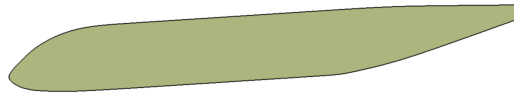


Figure 13: 2D Section without antenna

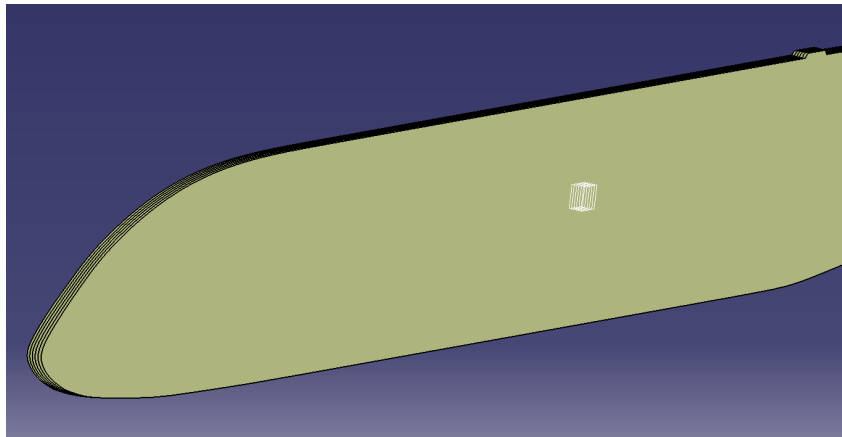


Figure 14: All section with antenna

Finally we export each of the sections in an igs format, this format preserve the geometry and avoid facetization occurs by STL format for example. This facetization is great for compressing model but in this case we want a raw model so igs is the best for our situation.

3.3 Fluid domain modeling

In any CFD analysis of aircraft antennas, the definition of the fluid domain plays a critical role in ensuring both the physical validity and numerical stability of the simulation. The fluid domain represents the virtual air volume surrounding the aircraft and antenna in which the governing Navier–Stokes equations are solved. If the computational domain is too small, artificial boundary effects can contaminate the flow field around the antenna, leading to inaccurate predictions of drag, lift, and pressure distribution. Conversely, an excessively large domain unnecessarily increases the total number of cells, thereby raising computational cost without improving accuracy.

Our fluid domain will be based on NACA airfoil wind-tunnel simulation :

- **Upstream distance:** The inlet boundary should be located around the upstream of the leading edge. This allows the inflow to fully develop before reaching the airfoil and avoids spurious pressure gradients at the inlet.
- **Downstream distance:** The outlet boundary must be placed significantly farther away of the trailing edge. This ensures that wakes and vortices are convected out of the domain without artificial reflection into the flow field.
- **Normal (vertical) distance:** The upper and lower boundaries are usually set at 3 times the length of the foil to minimize blockage effects and prevent confinement of shocks or vortices near the airfoil.
- **Far-field boundaries:** To mimic free-flight conditions, far-field boundaries should be defined with appropriate boundary conditions (e.g., pressure far-field with Mach number, static pressure, and angle of attack inputs).

Since we will use coordinated flow, we can remove farfield and only make our fluid domain with inlet and outlet. So here is a view of the fluid domain used for the simulation:

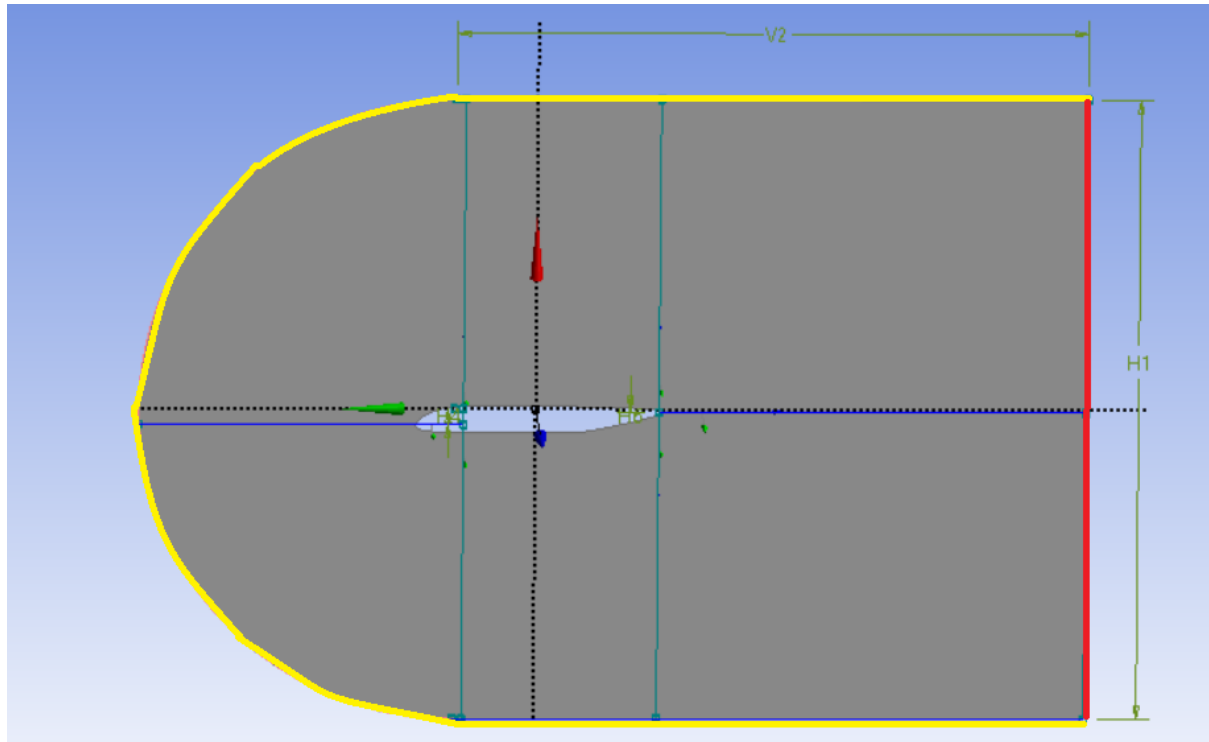


Figure 15: Fluid domain

The inlet is highlighted in yellow and the outlet in red. The blue lines have objective to create sector of our simulation to refine the mesh per sector and not all the fluid domain. This help the simulation to be more precize on critical zone (antenna).

3.4 Flight Conditions, Viscosity, Reynolds Number and Estimation of y^+

Context and chosen parameters. We first define the **flight conditions** of the aircraft under study, which will be the basis for all aerodynamic calculations: the **cruise velocity** (here 828 km/h, i.e. ≈ 230 m/s), the **characteristic length** (representative of the geometry; here $L = 38.67$ m), and the **altitude** (12 500 m). At this altitude, the chosen air density is $\rho = 0.33$ kg/m³. For viscosity, the operational value at the local temperature is used: $\mu = 1.320\,43e-5$ Pa s.

Step 1 — Checking the units before applying formulas. All formulas below assume *SI units*: ρ in kg/m³, U in m/s, L in m, μ in Pa s = kg/(m s). The conversion of 828 km/h to 230 m/s ensures consistency and prevents order-of-magnitude errors (particularly for viscosity).

Step 2 — Reynolds number (flow regime). The Reynolds number compares inertia to viscosity and indicates the flow regime (laminar/turbulent):

$$Re = \frac{\rho U L}{\mu}.$$

With $\rho = 0.33$ kg/m³, $U = 230$ m/s, $L = 38.67$ m, and $\mu = 1.320\,43e-5$ Pa s, the numerator is $\rho UL = 0.33 \times 230 \times 38.67 \approx 2.935 \times 10^3$, and

$$Re \approx \frac{2.935 \times 10^3}{1.32043 \times 10^{-5}} \approx \mathbf{2.22 \times 10^8}.$$

Such a Reynolds number confirms a **fully turbulent flow** at the aircraft scale.

Step 3 — Kinematic viscosity (preparing for y^+). For wall-related quantities we use the **kinematic viscosity** $\nu = \mu/\rho$. It represents the “viscous diffusivity per unit mass” and is expressed in m²/s. Here:

$$\nu = \frac{1.32043 \times 10^{-5}}{0.33} \simeq \mathbf{4.00 \times 10^{-5}} \text{ m}^2/\text{s}.$$

Step 4 — Estimation of wall shear stress and friction velocity u_τ . The parameter y^+ depends on the **friction velocity** u_τ , linked to the **mean wall shear stress** τ_w . Since this cannot be measured directly, we adopt the Prandtl–Schlichting correlation for a turbulent flat plate as an order-of-magnitude estimate of the **skin-friction coefficient**:

$$C_f = \frac{0.455}{(\log_{10} Re)^{2.58}}.$$

With $Re \approx 2.22 \times 10^8$, we obtain $C_f \approx \mathbf{1.91 \times 10^{-3}}$. The mean wall shear stress is then $\tau_w = \frac{1}{2}\rho U^2 C_f$, and the friction velocity

$$u_\tau = \sqrt{\frac{\tau_w}{\rho}}.$$

Numerically, this gives $\tau_w \approx 16.65$ Pa and $\boxed{u_\tau \approx 7.10 \text{ m/s}}$.

Step 5 — Definition and control of y^+ (first cell height). By definition,

$$y^+ = \frac{y u_\tau}{\nu} \implies y = \frac{y^+ \nu}{u_\tau}.$$

This expression allows setting a **target y^+** (e.g. 1 to resolve the viscous sublayer; 30–100 for wall-function RANS) and **deducing the physical first-cell height**. With $\nu \simeq 4.00 \times 10^{-5} \text{ m}^2/\text{s}$ and $u_\tau \simeq 7.10 \text{ m/s}$,

$$\frac{\nu}{u_\tau} \approx 5.63 \times 10^{-6} \text{ m}, \implies \boxed{y \approx y^+ \times 5.63 \times 10^{-6} \text{ m}}.$$

Examples for mesh sizing:

$$\begin{aligned} y^+ = 1 &\implies y \approx 5.6 \mu\text{m}, \\ y^+ = 30 &\implies y \approx \mathbf{0.17 \text{ mm}}, \\ y^+ = 100 &\implies y \approx \mathbf{0.56 \text{ mm}}. \end{aligned}$$

Remark on viscosity (Sutherland). If desired, μ can be derived from the local temperature using Sutherland's law: $\mu(T) = \mu_0(T/T_0)^{3/2} \frac{T_0+C}{T+C}$ (for dry air). At $T = 211.9 \text{ K}$, this yields $\mu_{\text{Suth}} \approx 1.40 \times 10^{-5} \text{ Pa}\cdot\text{s}$, about 5% higher than the chosen value. All subsequent results (Re , u_τ , y) would vary accordingly, but the orders of magnitude and meshing conclusions remain unchanged.

3.5 Definition of the mesh

Now that we determined the y^+ , we can establish a mesh that will be our base our simulation.

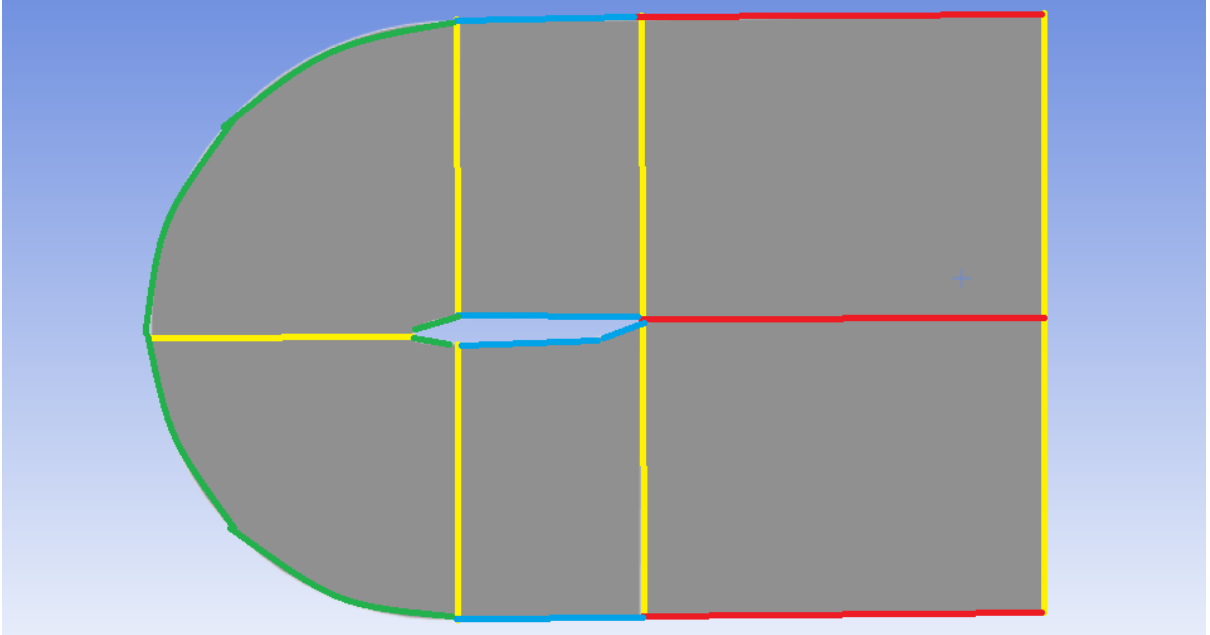


Figure 16: Meshing strategy

The mesh will be refine entirely with edge refinement, the main refinement is the normal refinement represented by the yellow line. This refinement will set the lenght of the

of each element at his normal to the airfoil. In order to limit the number of element but trying to match de y^+ , we will use a bias ratio that will increase the number of sections the more we are close to the airfoil. The 7 yellow edge will set a 300 divisions for each edge with a bias ratio of 100 000.

The red lines are for the horizontal refinement at the back of the airfoil, for the analysis of turbulence of a NACA model it is very usefull, in our case, since we only focus on fuselage and not stabilizer, we will set it with 200 divisions with no bias.

The blue lines set the main part of the fuselage with the antenna, this part will be divide in 150 divisions with no bias.

Finally the green lines representing the front part, here will be represented the first contact with the flow, it is set at 100 divisions for the 4 lines with no bias.

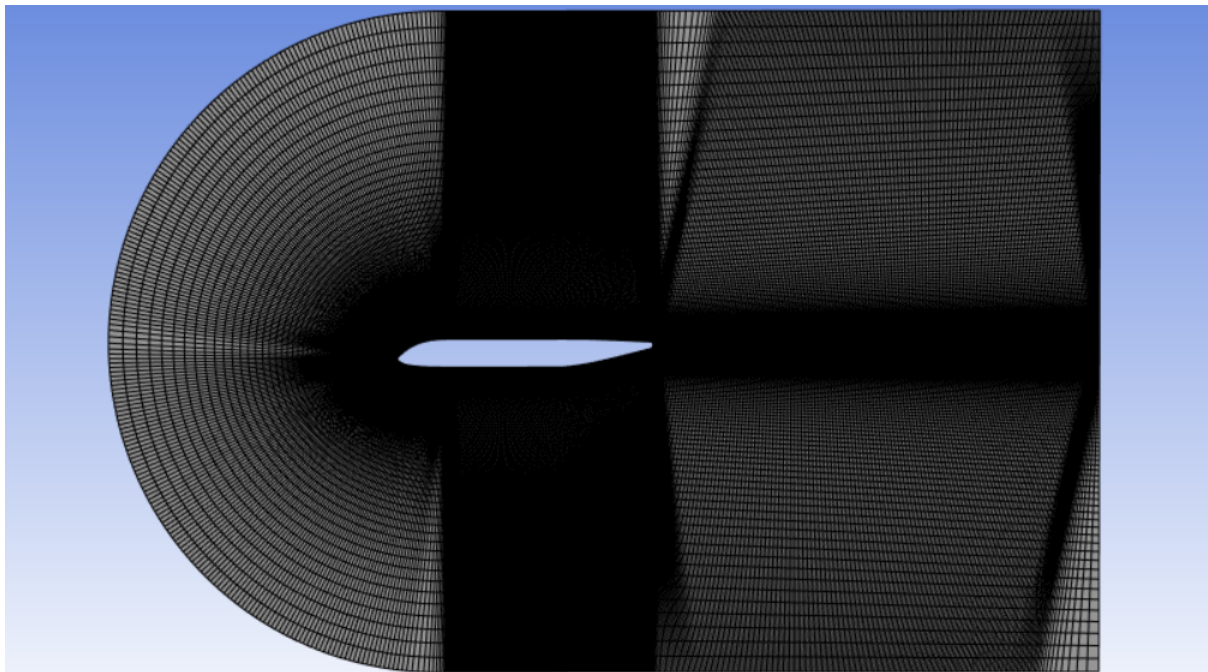


Figure 17: Final mesh

Here we can see the final mesh for all the analysis. We can clearly see the impact of the bias that focus the meshing around the airfoil.

4 Analysis and discussion

The results presented in this section are based on numerical simulations performed with *ANSYS Fluent*. The objective is to quantify the aerodynamic loads generated by a fuselage-mounted Ka-band antenna and to relate the computational outcomes with the theoretical framework introduced earlier, particularly regarding Reynolds number dependence, viscosity, and boundary layer behavior.

By comparing CFD results with these theoretical expectations, the analysis highlights how viscous effects, flow transition, and boundary-layer thickness interact with the antenna geometry. This approach also enables validation of whether the meshing strategy (with $y^+ \approx 1$) has correctly captured the near-wall physics, ensuring that viscous sublayer effects are resolved and that drag predictions remain reliable.

The following subsections present pressure distributions, drag coefficients, and flow visualizations obtained from Fluent, which are then interpreted in light of the Reynolds number calculations and atmospheric conditions. This will be presented Sections by Sections and a comparative work will be made for comparison.

4.1 Setup

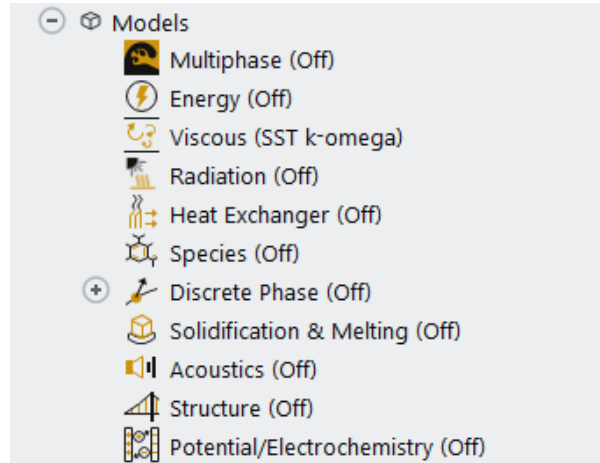


Figure 18: Models settings

In the present study, the *Shear Stress Transport (SST) $k-\omega$* turbulence model was adopted in *ANSYS Fluent* for the simulation of aerodynamic loads on the fuselage-mounted Ka-band antenna. The choice of this model is justified by several factors directly relevant to external aerodynamics at high Reynolds numbers:

- **Near-wall accuracy:** The SST $k-\omega$ model is formulated to provide robust predictions in the near-wall region, particularly when the mesh resolves the viscous sublayer with $y^+ \approx 1$. Since accurate representation of the boundary layer development around the antenna is critical to capture drag and flow separation, this feature makes it especially suitable.
- **Treatment of adverse pressure gradients:** Fuselage-mounted antennas introduce local flow acceleration followed by deceleration, resulting in strong adverse

pressure gradients. The SST $k-\omega$ model is known to perform reliably in these conditions, giving more accurate separation onset and wake predictions than the standard $k-\varepsilon$ models.

- **Blending advantages:** The SST formulation combines the $k-\omega$ model near the wall with the $k-\varepsilon$ formulation in the free stream. This hybrid approach avoids sensitivity to free-stream turbulence values while maintaining accuracy in the boundary layer, a crucial compromise for aircraft-scale flows.
- **Proven use in aerospace applications:** The SST $k-\omega$ model is widely used in the aerospace industry for aerodynamic protuberances, airfoils, and turbomachinery flows. It has been validated against experimental data for predicting drag, pressure distributions, and separation under transonic and subsonic flow regimes, all of which are relevant to the Ka-band radome configuration at cruise Mach numbers.

In summary, the SST $k-\omega$ model provides the necessary balance between computational efficiency and physical fidelity. Its ability to resolve near-wall effects while accurately predicting separated regions ensures reliable aerodynamic load estimation for the antenna-fuselage system.



Figure 19: Materials

Figure 20: Air definition

We define the materials as defined in the flight conditions section.

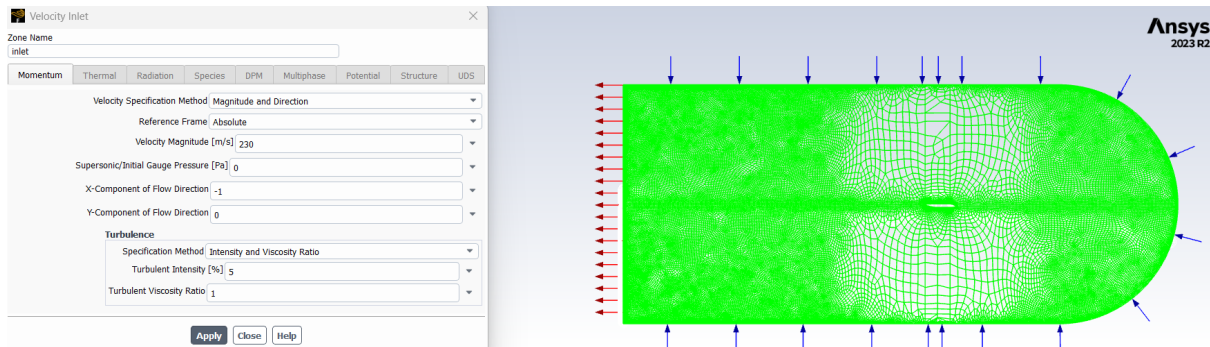


Figure 21: Inlet definition

Due to a little issue of reference plane, the simulation is set on -x. So the inlet direction is set at -1 on x component. Turbulent velocity ration is also set at 1 with the objective to simplify calculation and help to the convergence of the simulation.

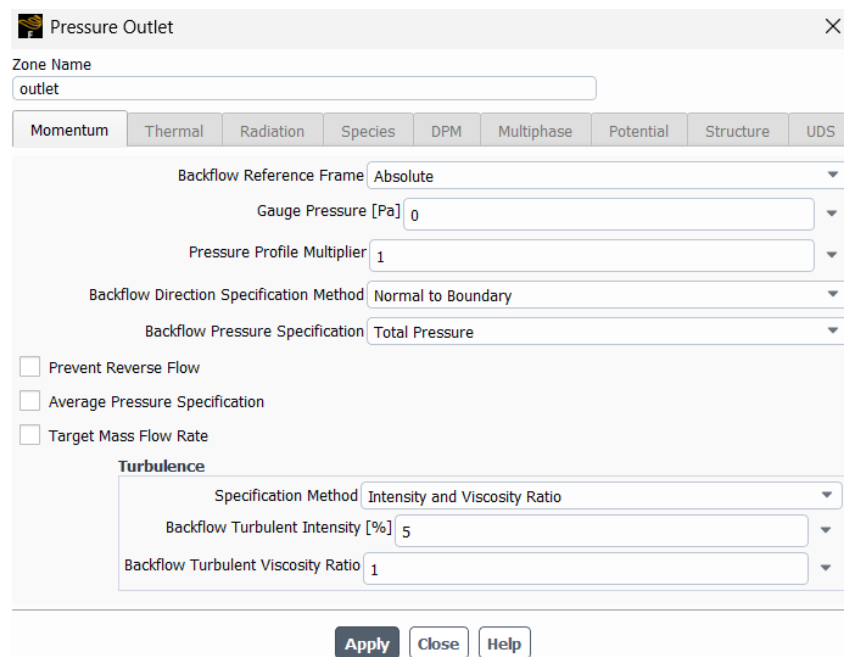


Figure 22: Outlet definition

Same settings for outlet, removing the turbulence viscosity ratio for helping convergence.

Other settings are set to default.

4.2 Solution

In order to have a view on the simulation, we create a drag and lift curve with the objective to see when the model converge. In fact, the flow starting in a turbulent view, this is translated by a drag and lift curve that have a lot of high and low and then when the model converge, the curve flatten and the model converge.

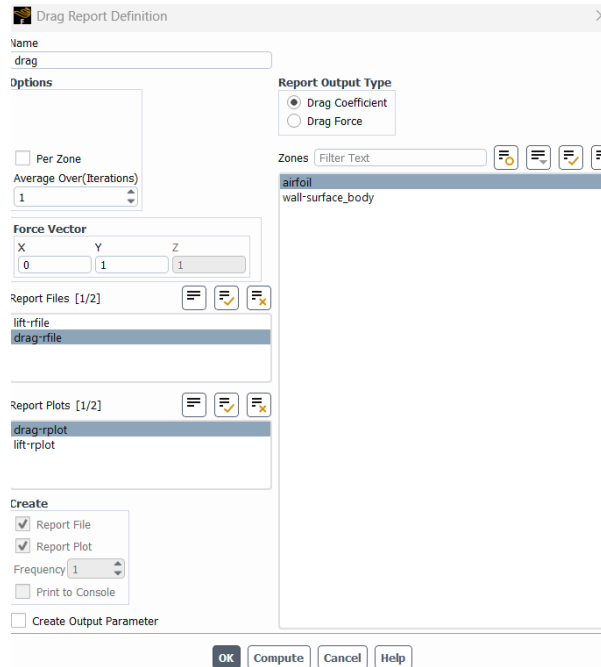


Figure 23: Drag curve settings

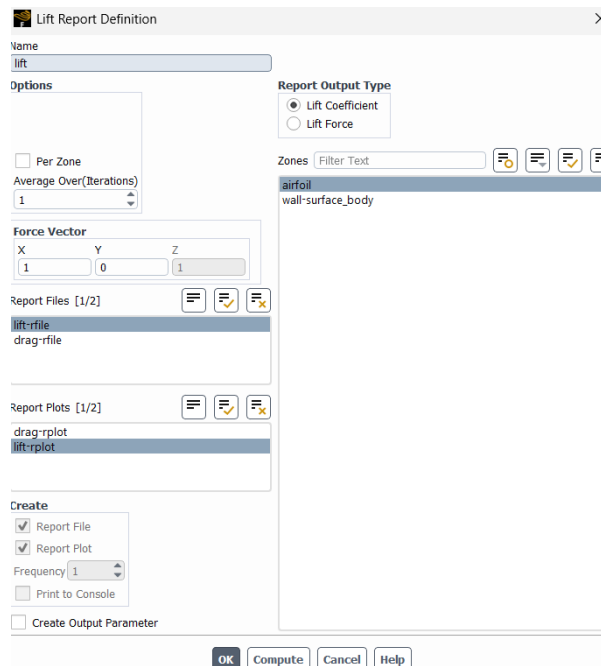


Figure 24: Lift curve settings

Finally the parameters for the calculations, i set every simulation at 1000 iterations

with an interval of 10 iterations, so the computer will calculate for 10 iterations and write down the results, it helps to have cleaner curves

4.3 Results

For this last part we will discover all the results obtained from the different simulations. We will compare these different sections following 3 countours analysis.

- Pressure coefficient C_p to locate any overpression or depression caused by the antenna that could result on more drag onto the fuselage.
- Velocity that determine a lot the drag resistance of the installation
- And finally the turbulence that find the potential critical point that can enter in vibration and cause lot of fatigue of the structure

All the discussed simulation are available on the Appendices section.

4.3.1 Section 0 — Comparison (0A = with antenna, 0NA = without)

Pressure coefficient C_p

In 0NA, the pressure field recovers smoothly along the body; the lowest C_p appears near mid-chord and rises gently toward the trailing end, with no localized spikes. In 0A, the fairing produces a sharp suction pocket on the forward shoulder followed by a rapid pressure rise over the aft fillet. This abrupt recovery establishes a strong adverse gradient consistent with a short separation bubble and leaves a localized high-pressure footprint at the trailing fillet. Overall, 0A introduces a marked C_p non-uniformity and a clear form-drag contribution that is absent in 0NA.

Velocity magnitude

Without the antenna (0NA) the outer flow remains nearly uniform and the boundary layer is thin and attached. With the antenna installed (0A) the flow accelerates over the crest and then decelerates into a low-velocity region immediately downstream; the wake persists for several antenna heights before full recovery. This momentum deficit corroborates the C_p behavior and reflects installation-induced losses.

Turbulent kinetic energy k

Case 0NA confines k to a narrow near-wall band typical of attached flow. Case 0A exhibits a compact k hotspot just aft of the trailing fillet, aligned with the separated shear layer, with elevated levels persisting downstream. The added mixing is the signature of separation/reattachment and implies higher broadband noise and potential vibration risk near this section.

4.3.2 Section 50 — Comparison (50A = with antenna, 50NA = without)

Pressure coefficient C_p

In the baseline 50NA, C_p varies smoothly along the body with a gentle suction over the upper surface and a regular pressure recovery toward the trailing end. When the antenna is installed (50A), the local C_p field shows a mild suction pocket on the forward shoulder of the fairing and a localized pressure rise over the aft fillet. The gradient is less abrupt

than at Section 0, and the footprint of the recovery remains compact; any separation bubble, if present, is short. Net effect: 50A introduces a localized form-drag contribution relative to 50NA, but it is weaker than at Section 0.

Velocity magnitude

Without the antenna (50NA) the outer flow remains uniform and the boundary layer stays thin and attached. With the antenna (50A) the flow accelerates over the crest and then decelerates into a shallow wake immediately downstream of the aft fillet. The wake thickness and recovery length are both modest compared with Section 0, indicating milder installation losses at this location. The qualitative trends in velocity and C_p are consistent.

Turbulent kinetic energy k

Case 50NA confines k to a narrow near-wall band typical of attached flow. Case 50A exhibits a compact k hot spot just aft of the trailing fillet, aligned with the separated shear layer; elevated levels decay within a short downstream distance. Compared with Section 0, the turbulence augmentation is smaller, suggesting reduced mixing losses and lower risk of broadband noise or vibration at Section 0.

Link to Section 0

The same installation mechanisms seen at Section 0 pressure rise over the aft fairing, a momentum deficit, and a k peak—are present here but noticeably attenuated. This indicates a smaller drag increment at Section 50 than at Section 0, either due to a gentler local contour or a better-aligned boundary layer at this section.

4.3.3 Section 100 — Comparison (100A = with antenna, 100NA = without)

Redundancy of Section 100. Section 100 (100A/100NA) repeats the same installation effects observed at Section 50: a mild suction pocket on the forward shoulder, a compact pressure rise at the aft fillet, a shallow wake with rapid recovery, and a small k hotspot. The signatures are slightly weaker than at Section 50 and introduce no new flow features (no additional separation or topology change). Consequently, Section 100 is redundant for the comparison and is cited only as “consistent with Section 50”; detailed figures are moved to the appendix for completeness.

4.3.4 Section 150 — Comparison (150A = with antenna, 150NA = without)

Pressure coefficient C_p .

In 150NA the C_p field is smooth with gentle suction along the profile and regular recovery at the trailing end. In 150A a localized suction pocket forms just upstream of the aft fillet and a compact pressure rise sits on the fillet; the recovery is confined and smoother than in earlier sections. No upstream disturbance is evident, indicating only a small added form-drag relative to 150NA.

Velocity magnitude.

Case 150NA shows attached flow with a thin boundary layer. In 150A the flow accelerates over the fairing crown and decelerates into a very shallow wake immediately downstream; recovery occurs within roughly one antenna height. The wake is thinner and shorter than in Section 50 and slightly weaker than in Section 100.

Turbulent kinetic energy k .

In 150NA, k remains confined to a narrow near-wall band. In 150A, a small k hotspot appears just aft of the fillet, aligned with the shear layer, and decays rapidly downstream. Its intensity is lower than Section 50 and comparable to or below Section 100, with no secondary hotspots.

4.3.5 Section 200 — Quantitative comparison (200A = with antenna, 200NA = without)

Velocity magnitude.

From the color scale, $U_\infty \approx 3.43 \times 10^2 \text{ m s}^{-1}$ (top of the legend). In **200A**, the immediate wake behind the aft fillet contains dark-blue to cyan bands, giving $U_{\min} \approx (3.5\text{--}7.0) \times 10^1 \text{ m s}^{-1} \approx 0.10\text{--}0.20 U_\infty$. By $x \approx 2h$ downstream, the wake recovers to the green-yellow bands, $U \approx (1.7\text{--}2.1) \times 10^2 \text{ m s}^{-1} \approx 0.50\text{--}0.62 U_\infty$, and to $0.9 U_\infty$ by roughly $x \approx 3\text{--}4h$. In **200NA** there is no comparable wake: outside the thin boundary layer the field stays at $2.06 \times 10^2 \text{ m s}^{-1}$ ($0.60 U_\infty$) over the same region.

Turbulent kinetic energy k .

The 200A legend spans up to $1.04 \times 10^3 \text{ m}^2 \text{ s}^{-2}$; the hotspot just aft of the fillet sits in the yellow band, $k_{\text{peak},A} \approx (5.2\text{--}6.3) \times 10^2 \text{ m}^2 \text{ s}^{-2}$. In 200NA the near-wall band peaks in the orange-yellow range, $k_{\text{peak},NA} \approx (2.3\text{--}3.7) \times 10^2 \text{ m}^2 \text{ s}^{-2}$. Thus the local amplification is $k_{\text{peak},A}/k_{\text{peak},NA} \approx 1.6\text{--}2.2$.

Pressure coefficient C_p .

The C_p legends cover roughly $[-1.9, -2.1] \times 10^4$ (blue) to $[1.3] \times 10^4$ (red). Both 200A and 200NA show similar suction minima on the lower surface ($C_p \approx -1.7 \times 10^4$). The installation effect in 200A is a compact positive patch at the aft fillet reaching the $[1.0\text{--}1.3] \times 10^4$ band, about 10–20% higher than the immediate upstream neighborhood; the footprint is h in streamwise extent.

Implication. At Section 200 the installation penalty is localized but measurable:

Table 1: Section 200: numerical summary from plot scales.

Metric	200A	200NA
U_∞ (from legend top)	$3.43 \times 10^2 \text{ m s}^{-1}$	$3.44 \times 10^2 \text{ m s}^{-1}$
$U_{\min}$ (wake)	$(3.5\text{--}7.0) \times 10^1 \text{ m s}^{-1}$	none (outside BL)
U at $x \approx 2h$	$(1.7\text{--}2.1) \times 10^2 \text{ m s}^{-1}$	$2.06 \times 10^2 \text{ m s}^{-1}$
k_{peak}	$(5.2\text{--}6.3) \times 10^2 \text{ m}^2 \text{ s}^{-2}$	$(2.3\text{--}3.7) \times 10^2 \text{ m}^2 \text{ s}^{-2}$
C_p (aft fillet max)	$(1.0\text{--}1.3) \times 10^4$	similar, slightly lower upstream

a wake with $U_{\min} \approx 0.1\text{--}0.2 U_\infty$, recovery to $0.5\text{--}0.6 U_\infty$ by $x \approx 2h$, and a $\sim (1.6\text{--}2.2) \times$ rise in peak k versus 200NA. The added drag is therefore dominated by a short, compact form-drag footprint with limited downstream influence.

4.3.6 Section 250 — Quantitative comparison (250A = with antenna, 250NA = without)

Velocity magnitude.

From the legend, the outer-field speed is $U_\infty \approx 3.56 \times 10^2 \text{ m s}^{-1}$. In **250A** the immediate wake behind the aft fillet shows dark-blue/cyan bands with $U_{\min} \approx (3.5\text{--}7.0) \times 10^1 \text{ m s}^{-1} \approx 0.10\text{--}0.20 U_\infty$. At $x \approx 2h$ downstream, the wake recovers to green-yellow bands, $U \approx (1.4\text{--}1.8) \times 10^2 \text{ m s}^{-1} \approx 0.40\text{--}0.50 U_\infty$, and to $0.9 U_\infty$ by $x \approx 3\text{--}4h$. In **250NA** there is no comparable wake: outside the boundary layer $2.14 \times 10^2 \text{ m s}^{-1}$ ($0.60 U_\infty$) over the same region. A visual read of the contours indicates a short separation bubble in 250A with reattachment within $L_r \approx 0.8\text{--}1.1 h$.

Turbulent kinetic energy k .

The 250A colorbar extends to $1.08 \times 10^3 \text{ m}^2 \text{ s}^{-2}$; the hotspot just aft of the fillet sits in orange-yellow bands, $k_{\text{peak},A} \approx (7.5\text{--}9.5) \times 10^2 \text{ m}^2 \text{ s}^{-2}$. In 250NA, the near-wall band peaks around yellow, $k_{\text{peak},NA} \approx (3.3\text{--}5.0) \times 10^2 \text{ m}^2 \text{ s}^{-2}$. Hence the local amplification is $k_{\text{peak},A}/k_{\text{peak},NA} \approx 1.5\text{--}2.3$.

Pressure coefficient C_p .

Both 250A and 250NA share similar suction levels on the lower surface ($C_p \approx -(1.5\text{--}1.9) \times 10^4$). The installation imprint in 250A is a compact positive patch at the aft fillet reaching the red band, $C_{p,\text{max},A} \approx (1.0\text{--}1.35) \times 10^4$, about 10–25% higher than the immediate upstream neighborhood. The streamwise footprint of this rise is short (h) and coincides with the separation/reattachment zone inferred above.

Table 2: Section 250: numerical summary from plot scales (A = with antenna; NA = without).

Metric	250A	250NA
U_∞ (legend top)	$3.56 \times 10^2 \text{ m s}^{-1}$	$3.56 \times 10^2 \text{ m s}^{-1}$
$U_{\min}$ (wake)	$(3.5\text{--}7.0) \times 10^1 \text{ m s}^{-1}$	none (outside BL)
U at $x \approx 2h$	$(1.4\text{--}1.8) \times 10^2 \text{ m s}^{-1}$	$2.14 \times 10^2 \text{ m s}^{-1}$
Reattachment length L_r/h	0.8–1.1	— (attached)
k_{peak}	$(7.5\text{--}9.5) \times 10^2 \text{ m}^2 \text{ s}^{-2}$	$(3.3\text{--}5.0) \times 10^2 \text{ m}^2 \text{ s}^{-2}$
$C_{p,\text{max}}$ (aft fillet)	$(1.0\text{--}1.35) \times 10^4$	lower, no localized rise

4.3.7 Boundary-layer behavior (overview across sections)

Upstream baseline (NA cases). Without the antenna the fuselage boundary layer is thin and fully attached in all sections; outer streamlines remain nearly parallel, C_p recovers smoothly, and k is confined to a narrow near-wall band—typical of a turbulent BL with shape factor $H \approx 1.3\text{--}1.4$.

Effect of the antenna (A cases). Installing the fairing thickens the boundary layer locally and, at the aft fillet, introduces a short separation bubble. In **Section 0** the BL interaction is strongest: the outer streamlines are visibly displaced away from the wall, a low-velocity near-wall core persists several h downstream, and the k hotspot marks a separated shear layer. Relative to 0NA, the *displacement thickness* δ^* is inferred to increase by roughly a factor of 2–3 over the bubble, with an effective shape factor peaking

above $H > 1.8$ (and locally much higher within the separated region where $C_f \rightarrow 0$).

Moving downstream to **Sections 50–150**, the bubble shortens and reattachment occurs closer to the fillet; the BL thickening is milder ($\delta_A^*/\delta_{NA}^* \approx 1.2\text{--}1.5$) and the k hotspot decays within $\sim 1\text{--}2h$. By **Sections 200–250**, the interaction is weakest: the separation length is short ($L_r \sim h$), recovery to $U_{0.9}U_\infty$ occurs within $3\text{--}4h$, and the BL parameters are close to the baseline ($\delta_A^*/\delta_{NA}^* \approx 1.1$). Across all A cases, the qualitative sequence is consistent: adverse gradient $\Rightarrow$ local C_f reduction/separation $\Rightarrow$ increased δ^* and wake deficit $\Rightarrow$ reattachment with renewed shear and gradual return toward the NA state.

4.4 Results conclusion

Across sections, the antenna installation introduces a localized adverse pressure gradient at the aft fillet that triggers a short separation bubble, a near-wall velocity deficit, and a k hotspot. The effect is strongest at Section 0 and attenuates downstream; by Sections 200–250 the bubble is short ($L_r \sim h$), the wake recovers within $3\text{--}4h$, and peak k rises by only about $1.5\text{--}2.3\times$ over the no-antenna case. Overall, the incremental drag is dominated by compact form-drag near the fairing with limited downstream influence, and Sections 50 (representative mid-strength) plus one of $\{200,250\}$ (representative mild) suffice for the main narrative.

5 Conclusion and Future Work

The present study investigated the aerodynamic behavior of a fuselage-mounted Ka-band antenna using a combined CAD/CFD approach. A simplified fuselage geometry with and without antenna fairing was created in CATIA V5, and multiple two-dimensional sections were analyzed in *ANSYS Fluent* with a high-fidelity meshing strategy ($y^+ \approx 1$) and the SST $k-\omega$ turbulence model.

The results confirmed that the antenna installation introduces localized aerodynamic disturbances primarily in the form of an adverse pressure gradient at the aft fillet. This gradient consistently generated a compact separation bubble, a momentum deficit in the near wake, and localized amplification of turbulent kinetic energy. The severity of these effects was found to be highest at the forward sections and progressively weaker downstream, with recovery of the flow field typically within three to four antenna heights. Comparisons against the baseline fuselage without antenna demonstrated that the incremental drag is dominated by compact form drag, with limited downstream influence. Overall, the installation penalty is non-negligible but spatially confined, highlighting the importance of fairing design in minimizing adverse aerodynamic effects.

Despite the simplifications introduced (2D sectioning, exclusion of wings and empennage, assumption of symmetric flow), the methodology has proven effective in capturing the essential aerodynamic mechanisms of antenna–fuselage interaction. The approach balances computational efficiency with physical fidelity, providing a valuable tool for preliminary assessment of external protuberances on transport aircraft.

5.1 Future Work

Several extensions of this study are recommended to build on the present findings:

- **Full 3D fuselage simulations:** A complete three-dimensional model including fuselage curvature, wing–body interaction, and empennage effects would capture spanwise flow phenomena neglected in the 2D section approach.
- **Shape optimization of the antenna fairing:** Parametric variations of length-to-height ratio, tail angle, and surface smoothness could be explored to minimize drag penalties while maintaining structural and RF performance.
- **Multiple flight conditions:** The present work focused on cruise at high altitude. Future studies should extend the analysis to additional operating points such as take-off rotation, climb, approach, and landing, where angles of attack and dynamic pressures differ significantly. These regimes may represent more critical conditions for flow separation, buffeting, and structural loading on the antenna installation.
- **Unsteady simulations (URANS or DES):** Transient flow analyses would allow evaluation of broadband noise, buffeting, and vibration risks associated with unsteady separation behind the fairing.
- **Experimental validation:** Wind-tunnel tests or scaled models could provide pressure coefficient and drag data for comparison, strengthening confidence in the numerical predictions.

- **Multi-disciplinary integration:** Coupling the aerodynamic loads with structural finite-element analysis would enable a comprehensive assessment of antenna mounting stresses and potential fatigue issues.
- **Alternative antenna concepts:** Finally, conformal or flush-mounted antenna concepts, as explored in projects such as ACASIAS, should be studied as a long-term solution to eliminate external drag penalties altogether.

5.2 Personal conclusion

Beyond the technical outcomes, this project has been a highly valuable learning experience. It allowed me to deepen my understanding of Computational Fluid Dynamics within the context of aerospace applications, particularly in the modeling of complex flows around fuselage-mounted installations. By working through the full process, from CAD geometry preparation to mesh design, solver setup, and interpretation of aerodynamic results, I gained practical insight into both the capabilities and limitations of CFD tools.

On a personal level, this study has significantly strengthened my confidence in conducting aerodynamic analyses and has improved my ability to critically assess numerical results against theoretical expectations. On a professional level, it represents an important step in developing the skills required for engineering practice in the aerospace sector. I believe the knowledge and methodology acquired here will be directly transferable to future projects, making this thesis not only a piece of academic work but also a long-term asset for my career development.

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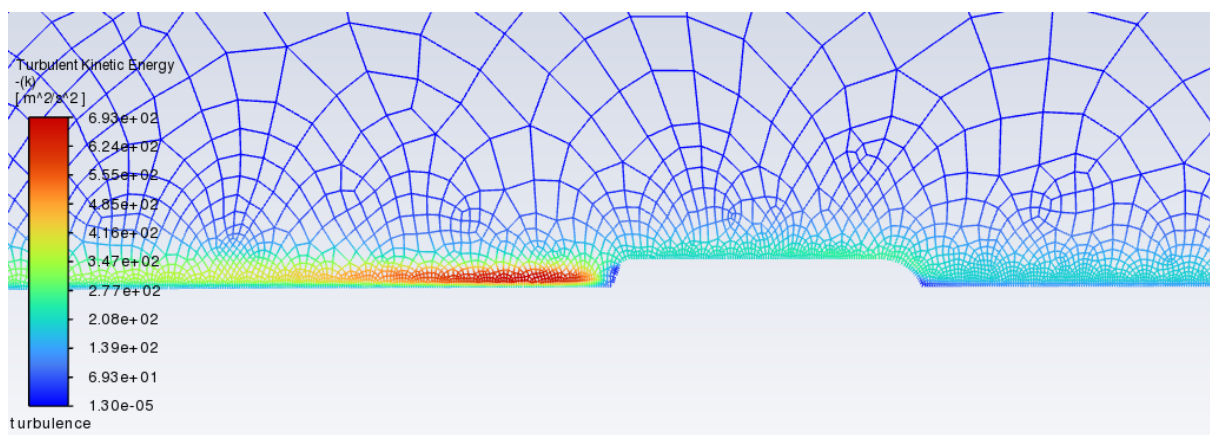
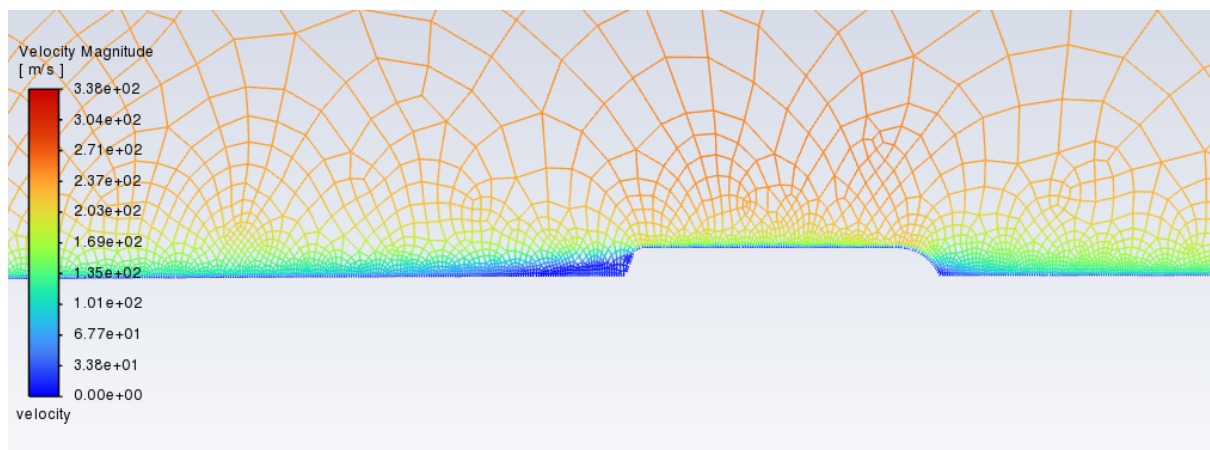
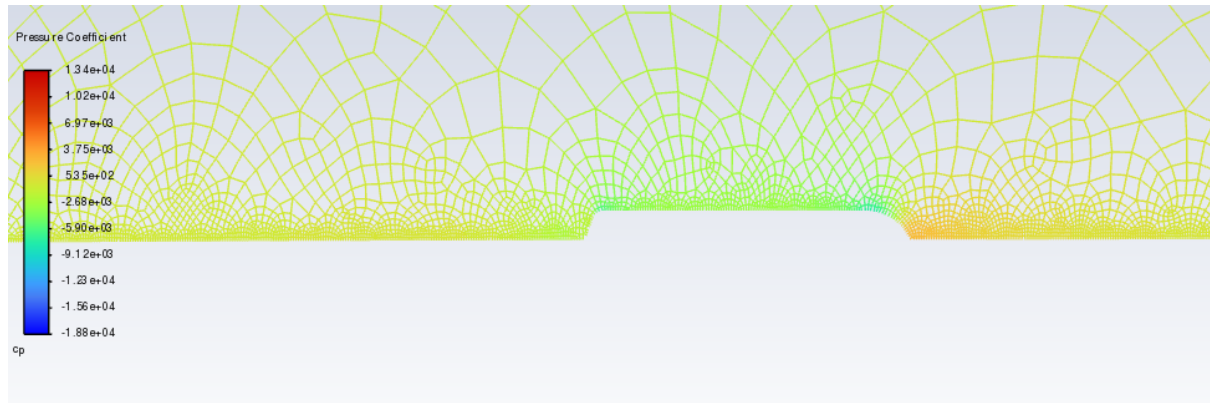
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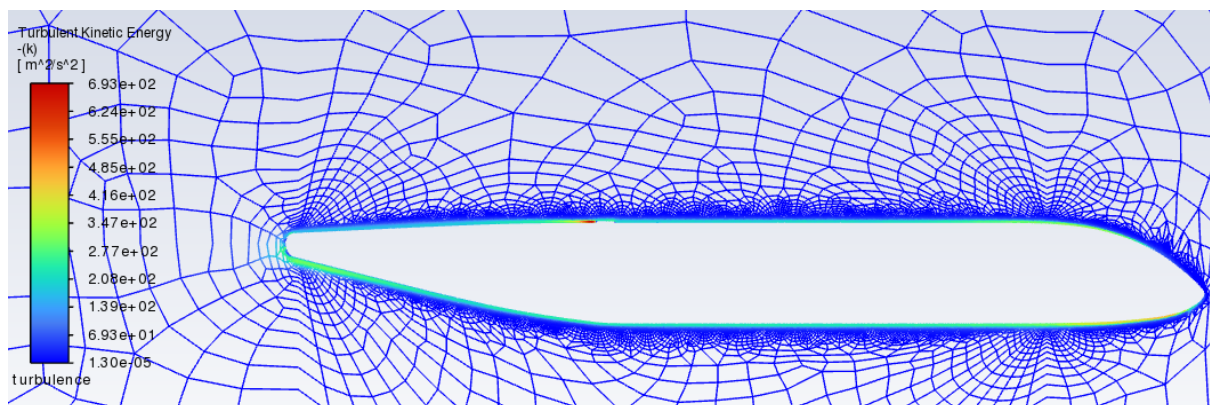
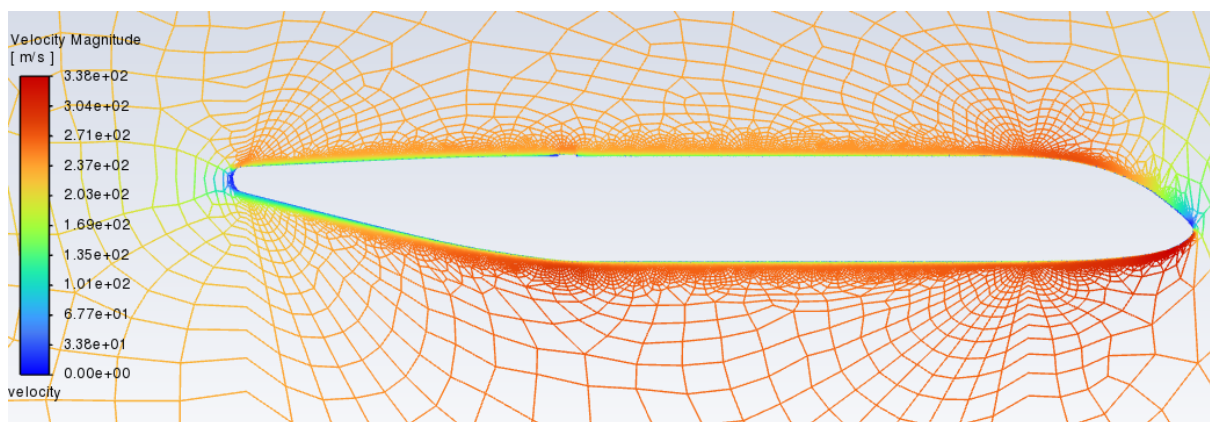
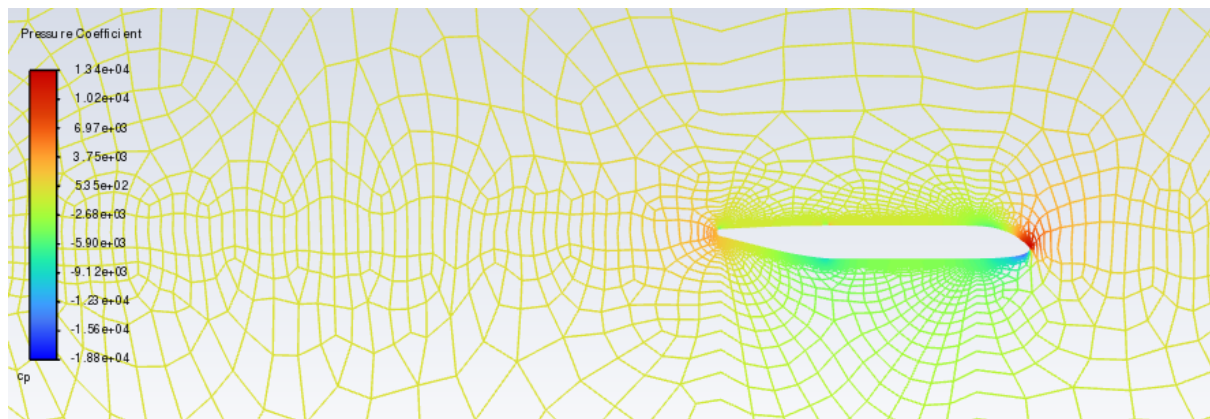
Appendices

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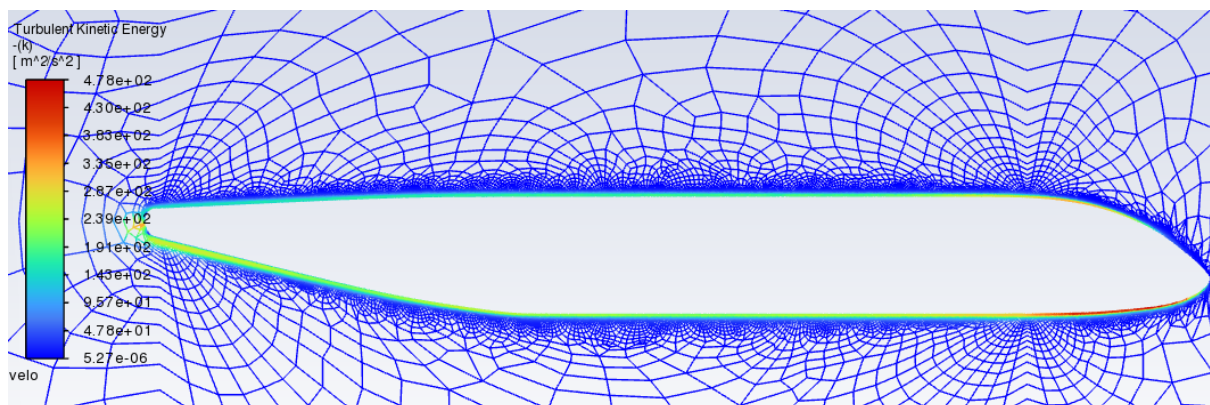
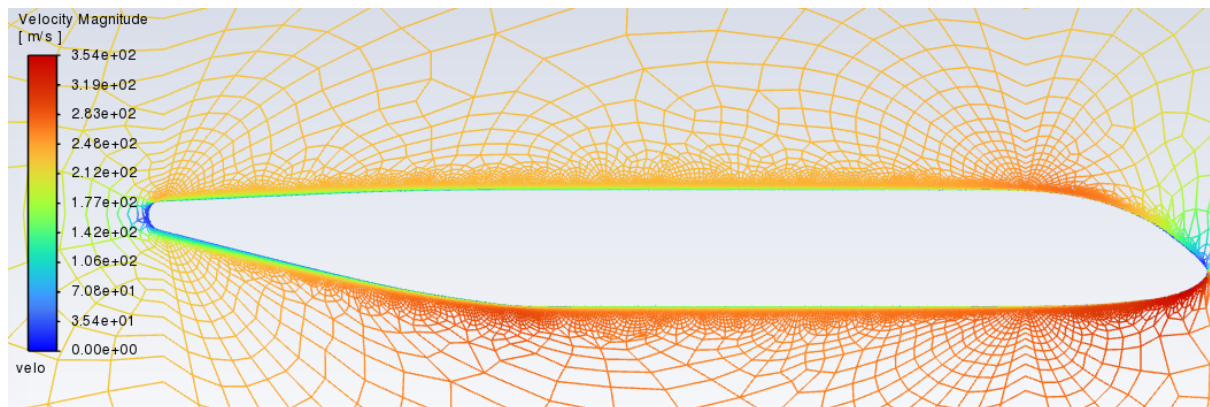
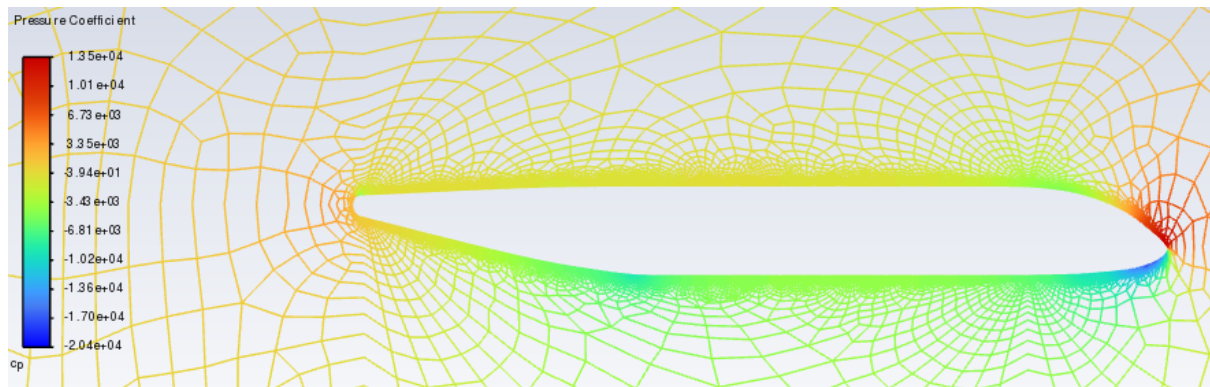
Always on Cp - Velocity - Turbulence order

.1.1 Antenna



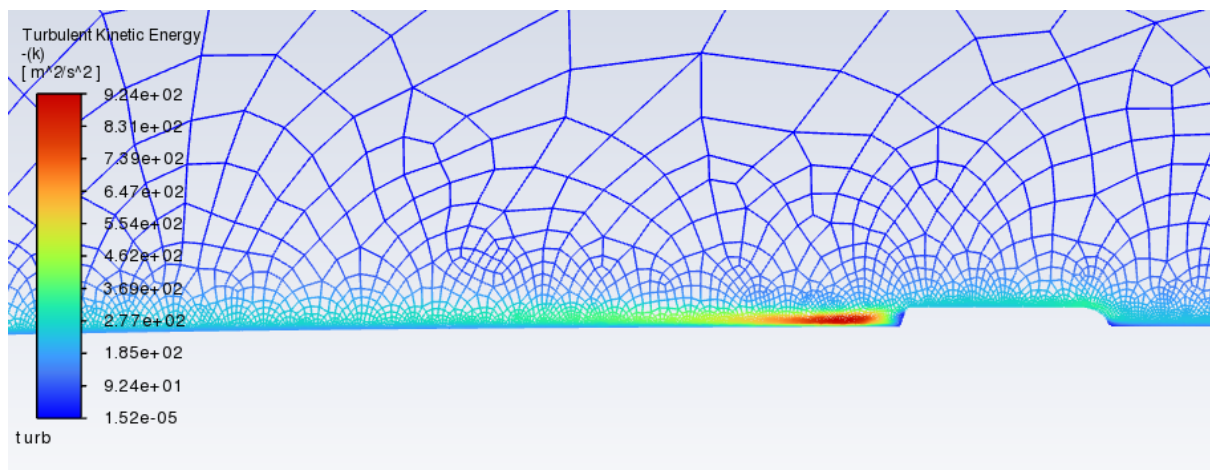
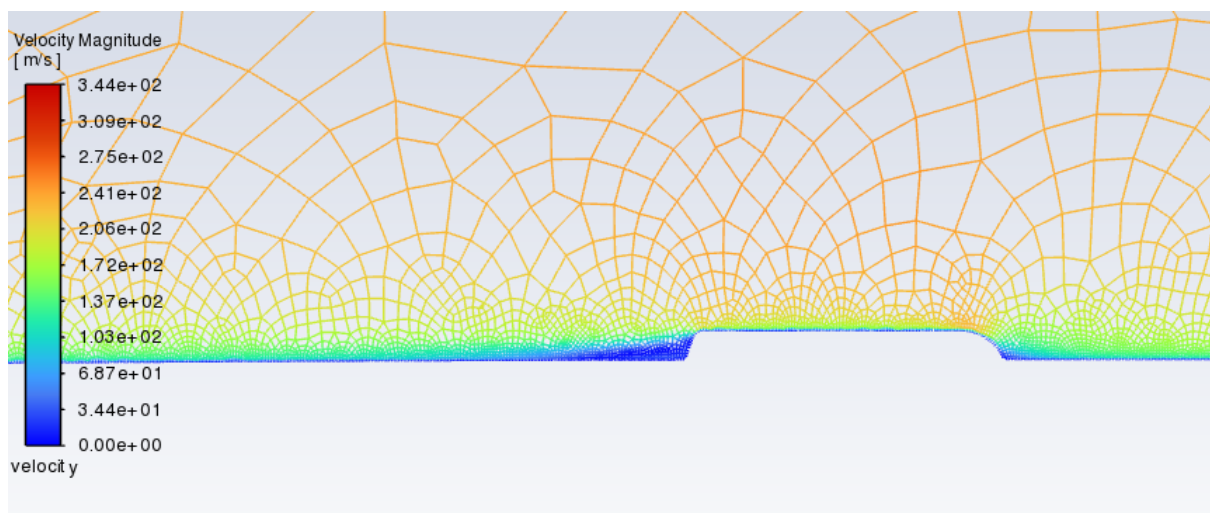
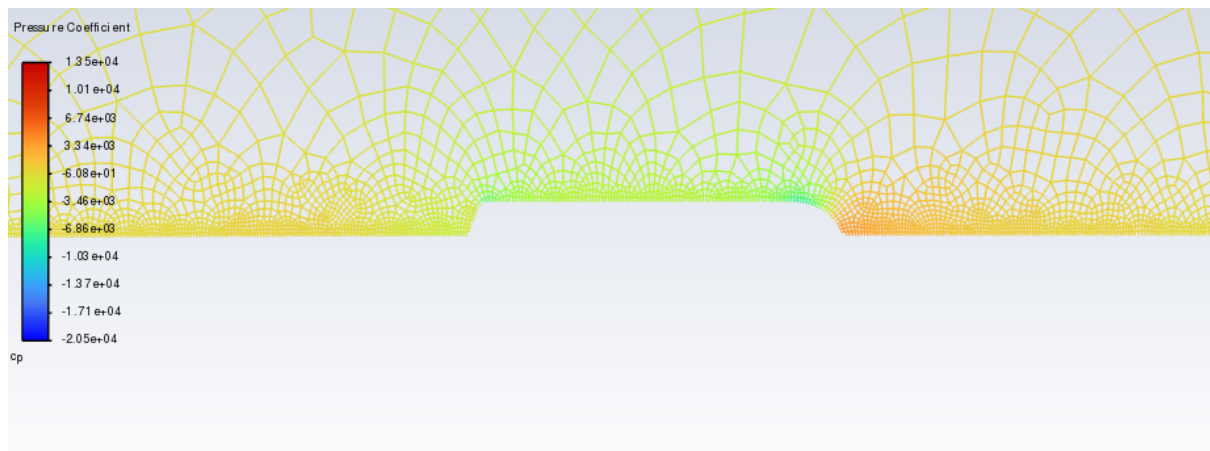


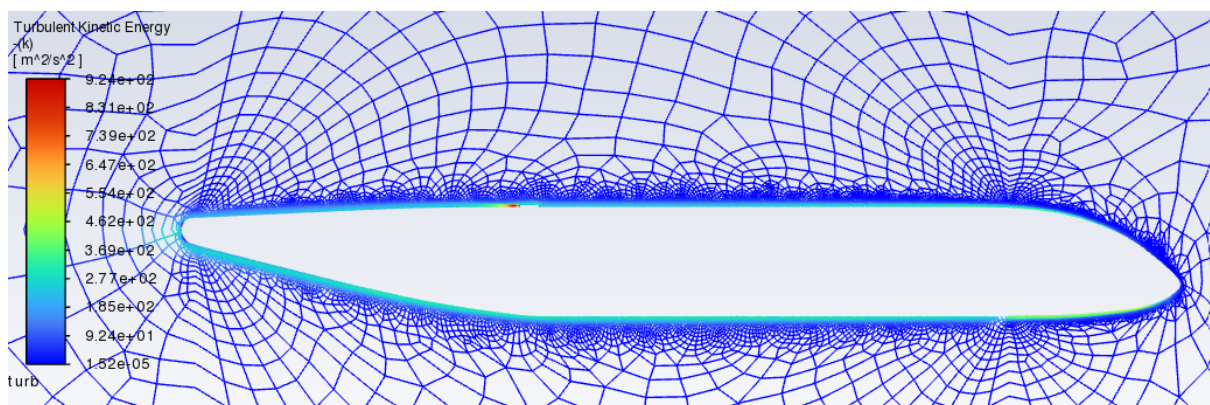
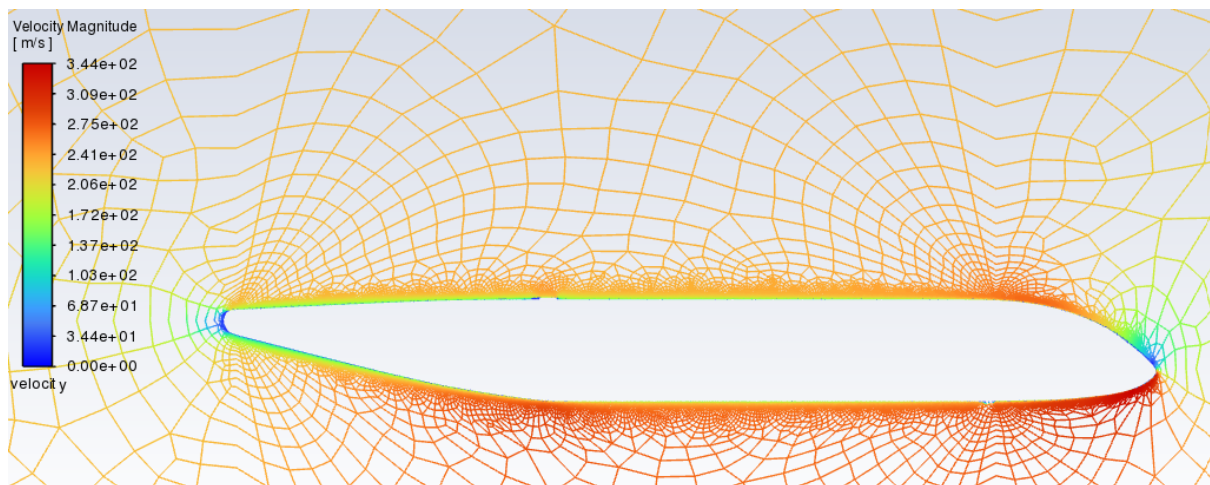
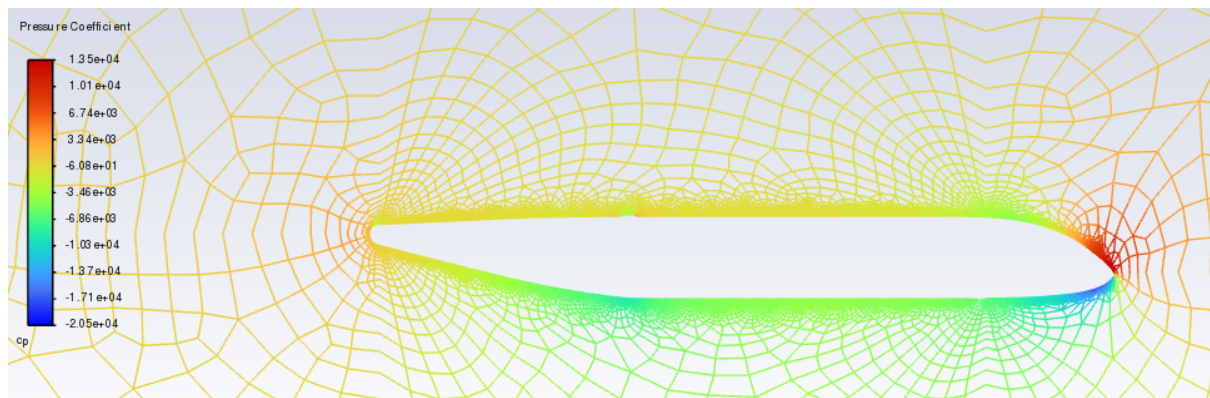
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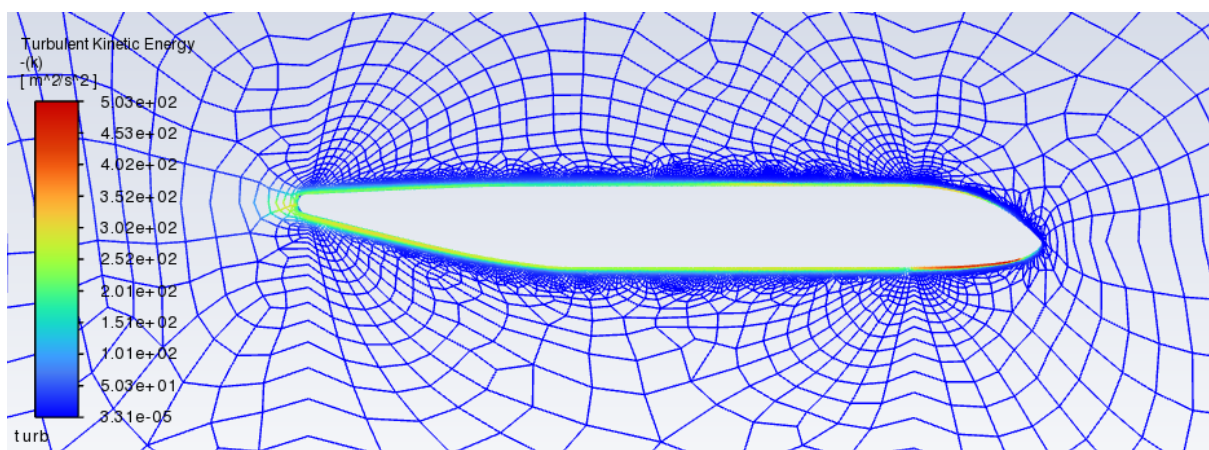
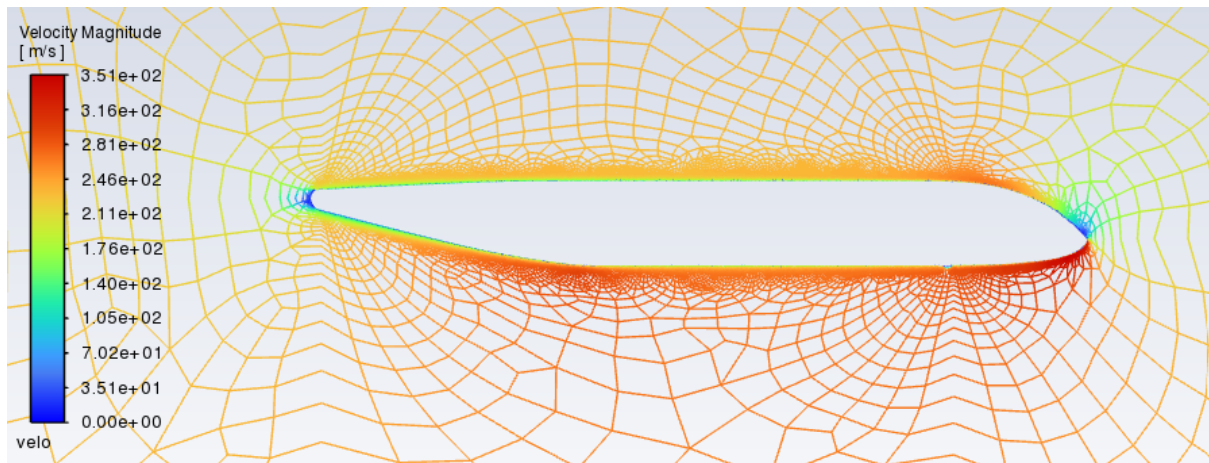
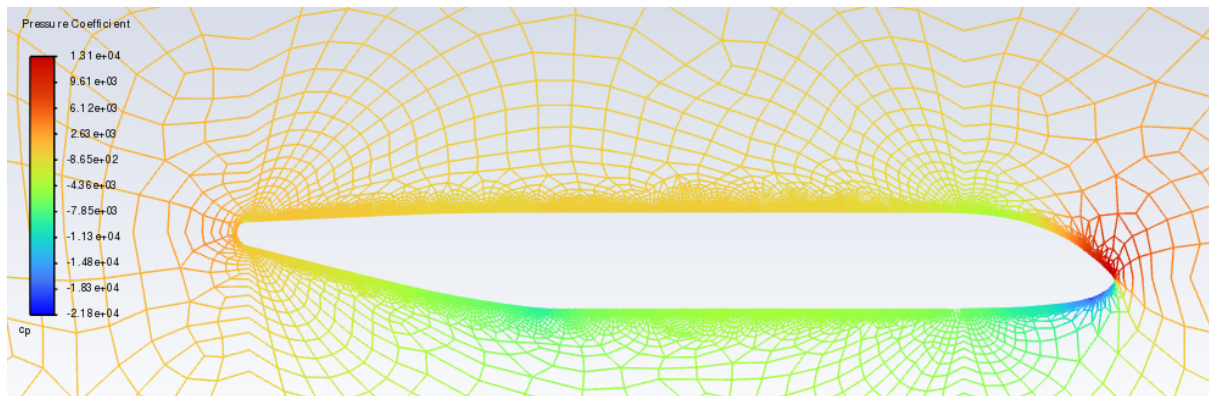
.2 Section 50

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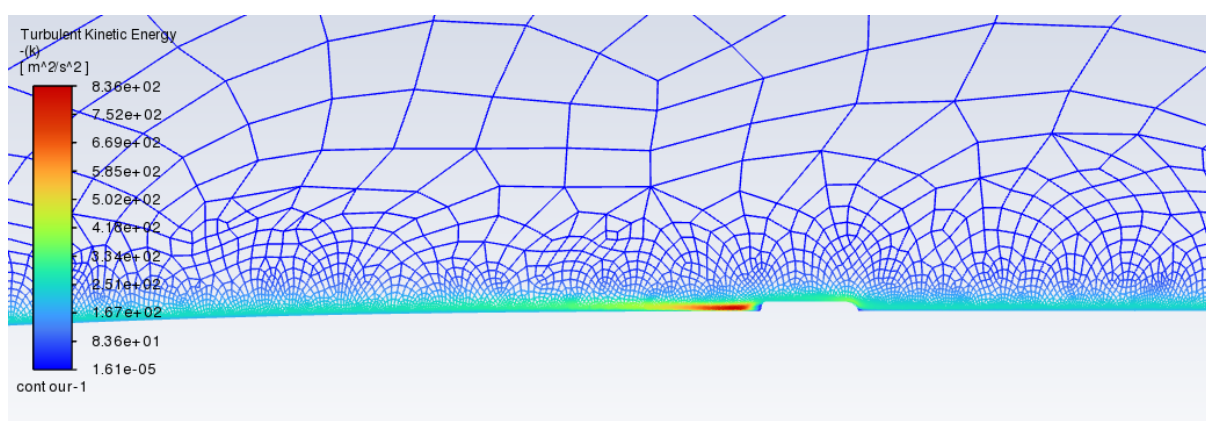
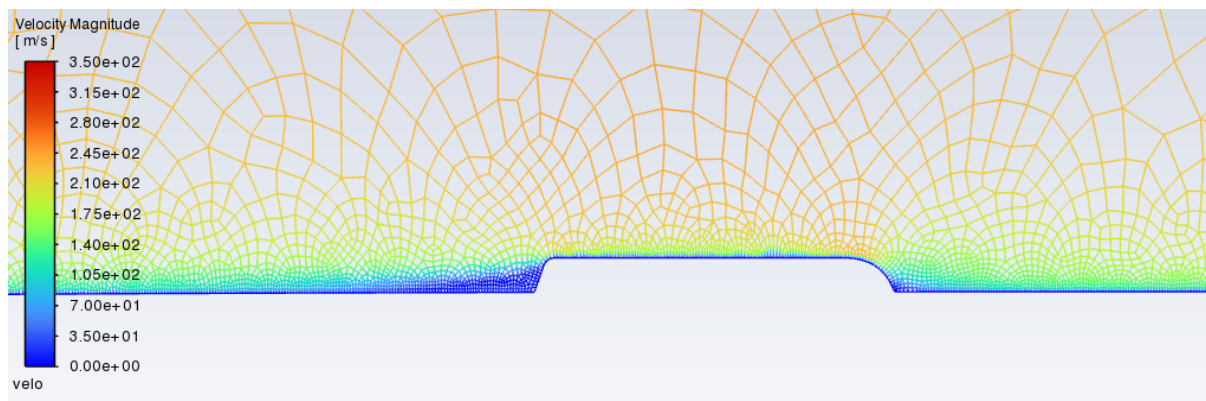
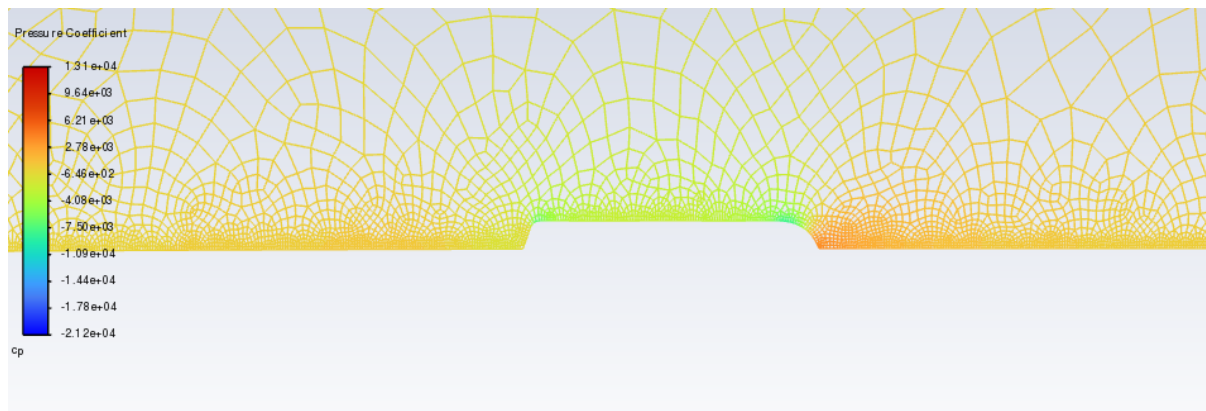


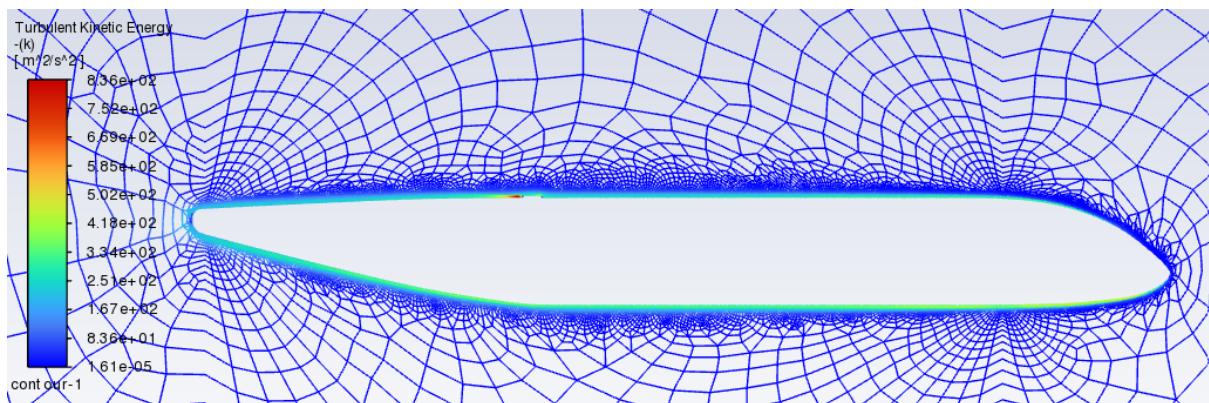
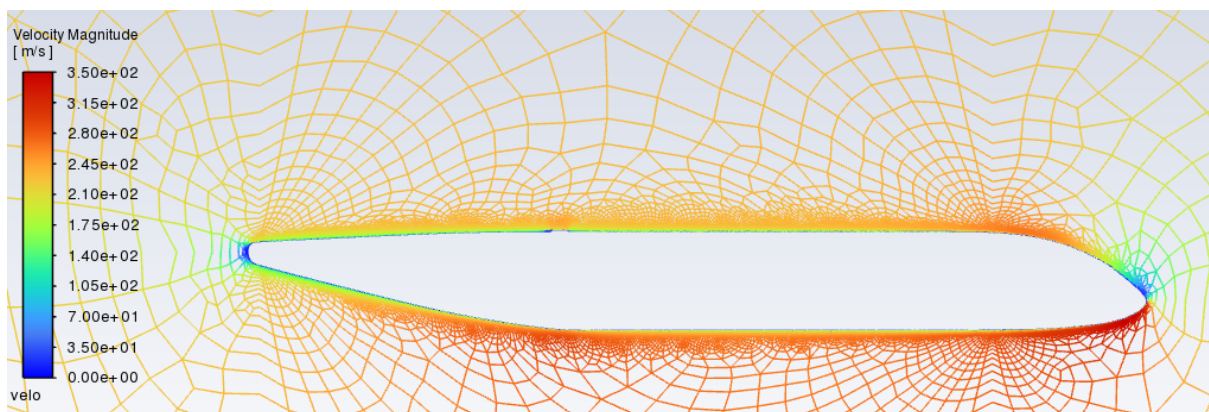
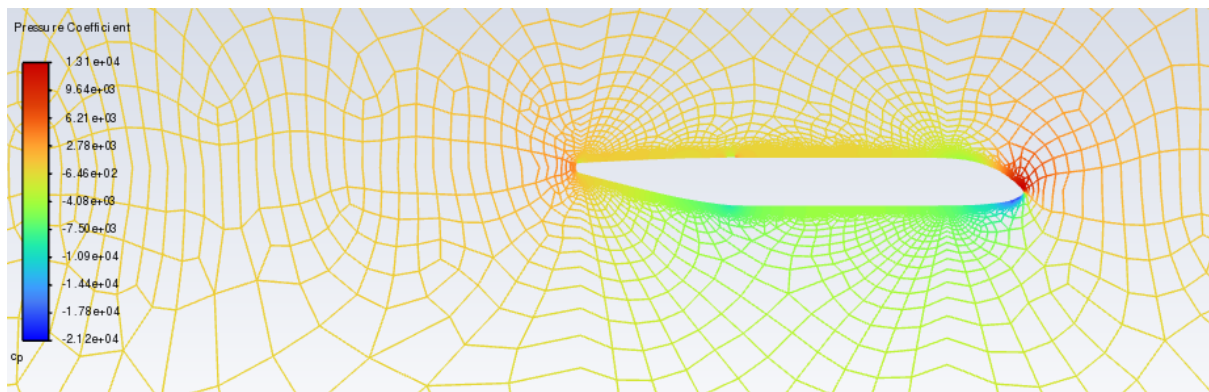
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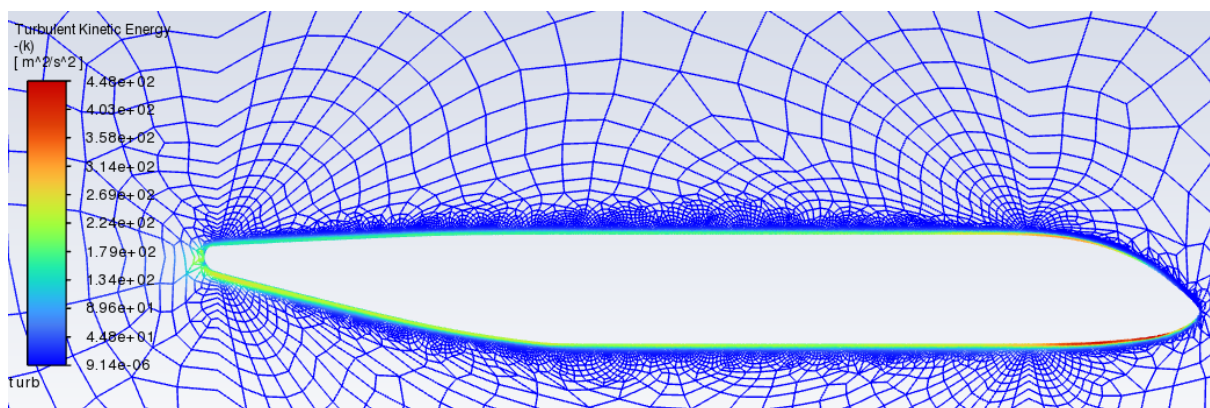
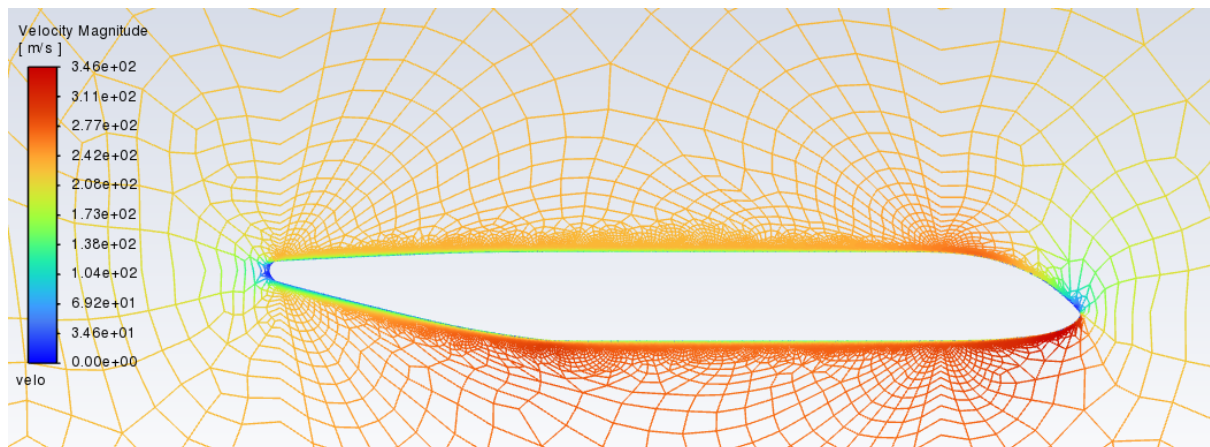
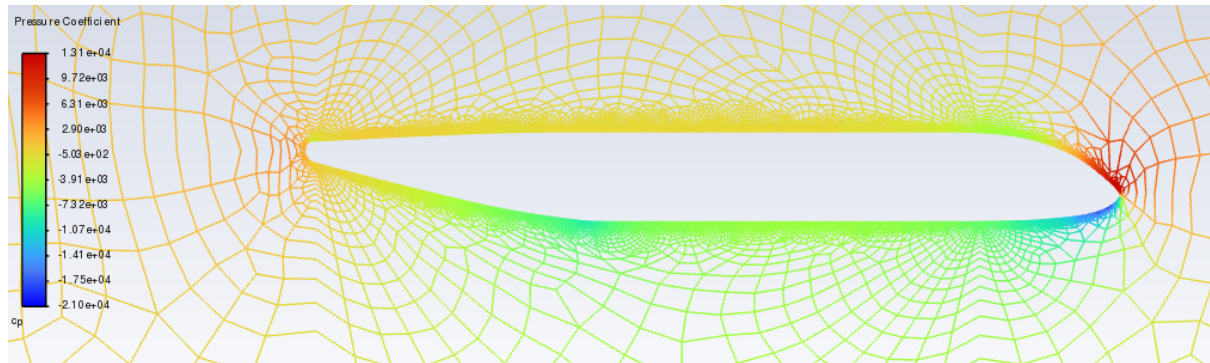
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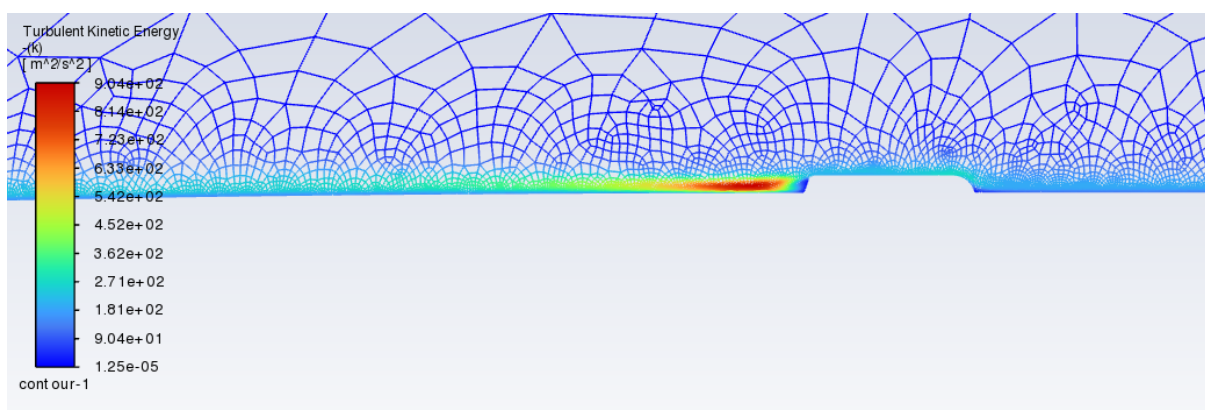
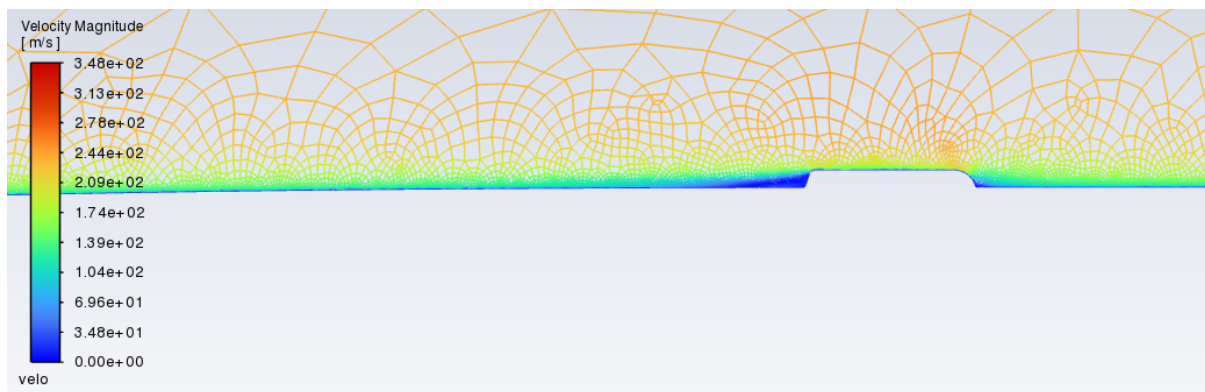


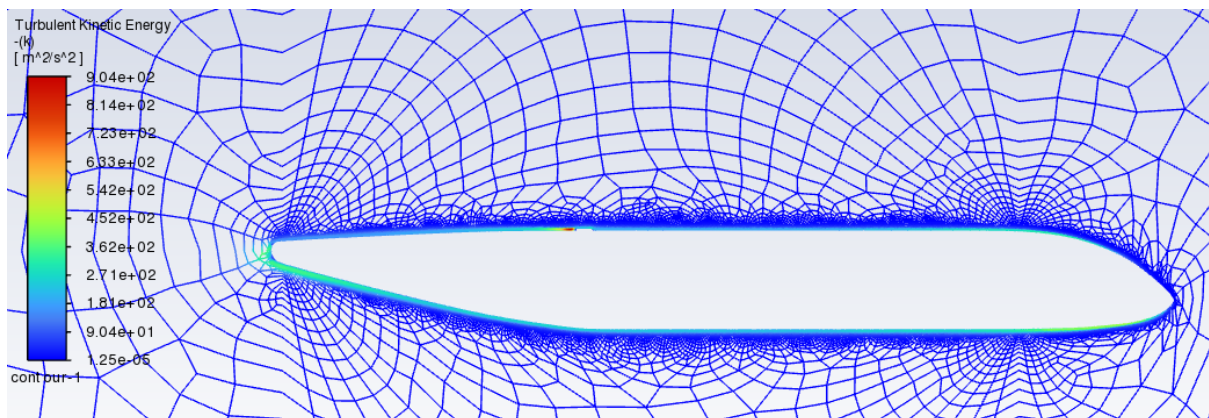
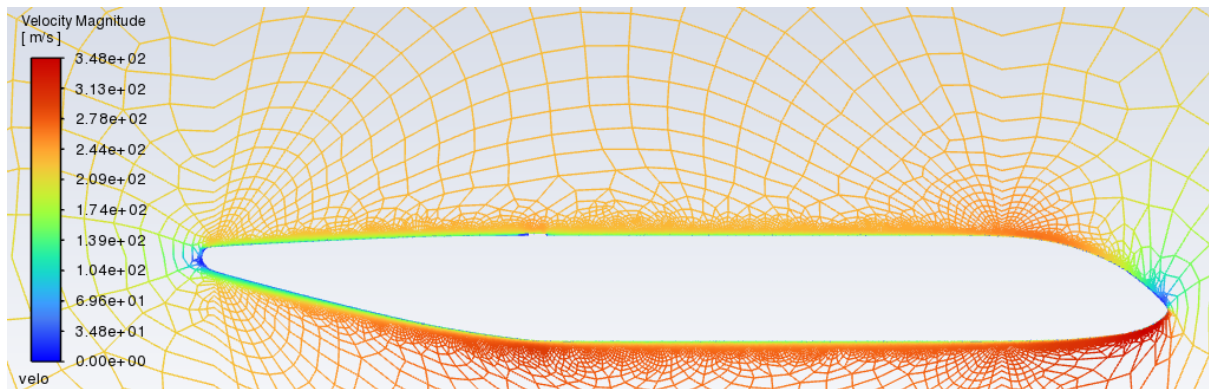
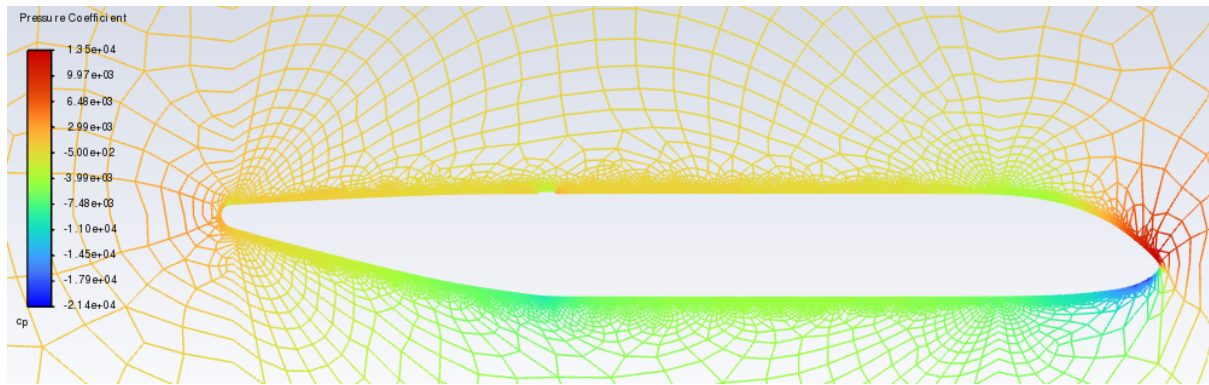
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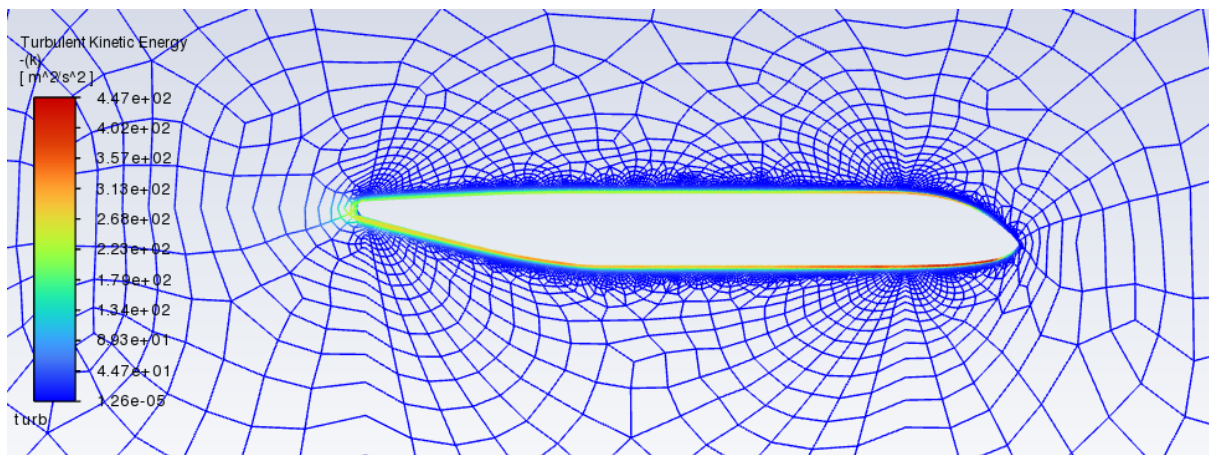
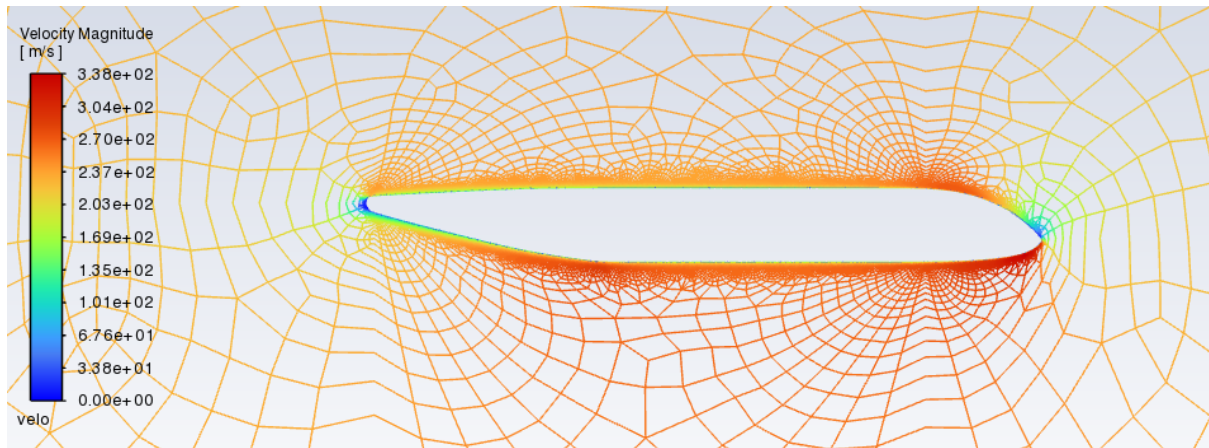
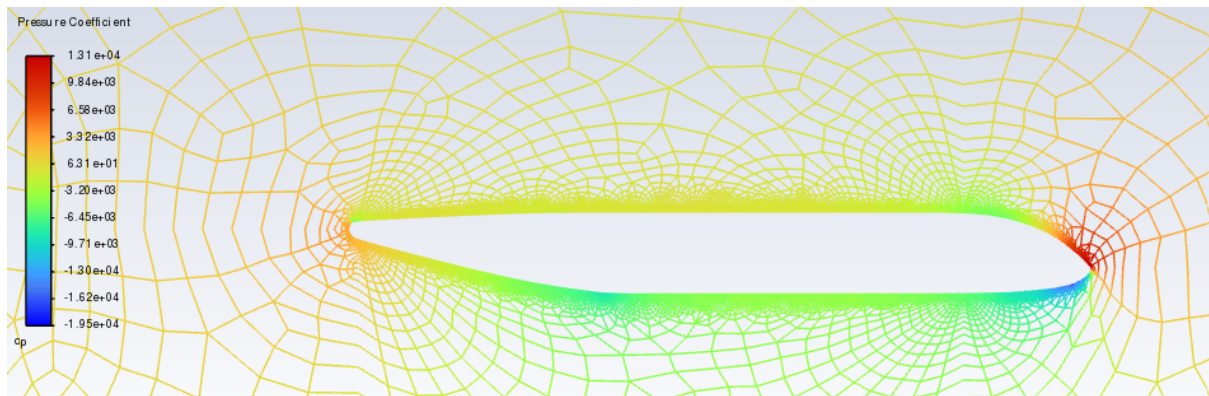
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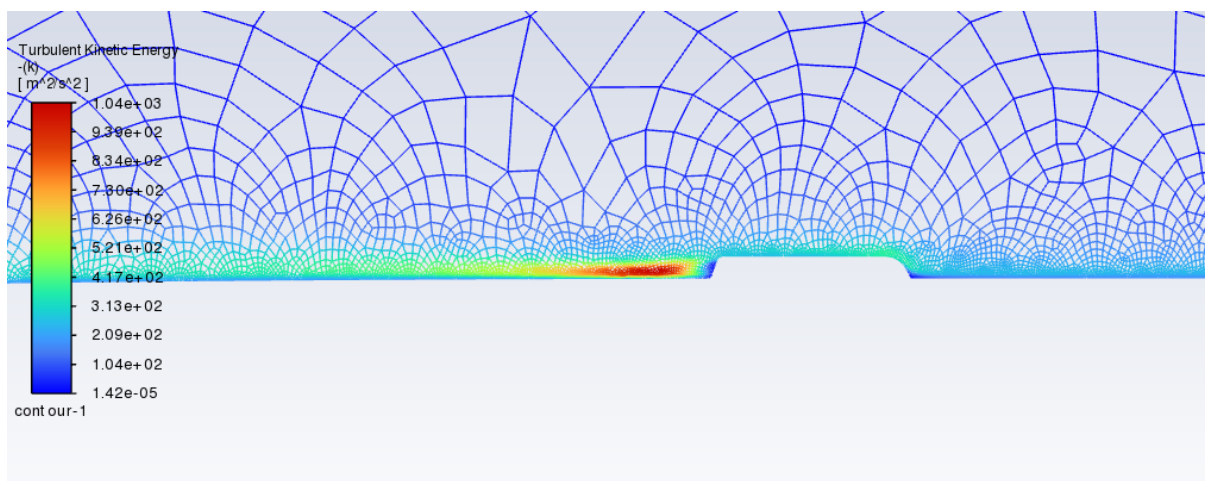
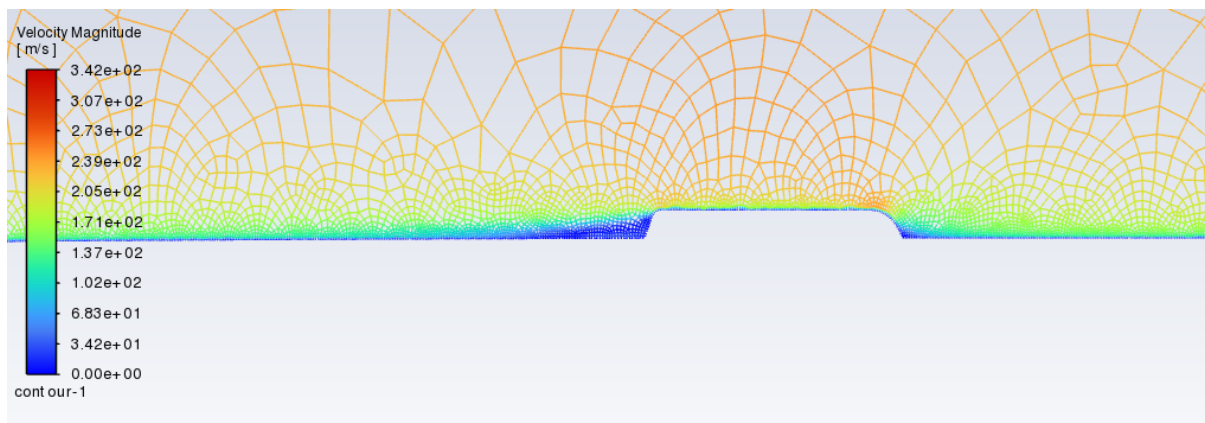
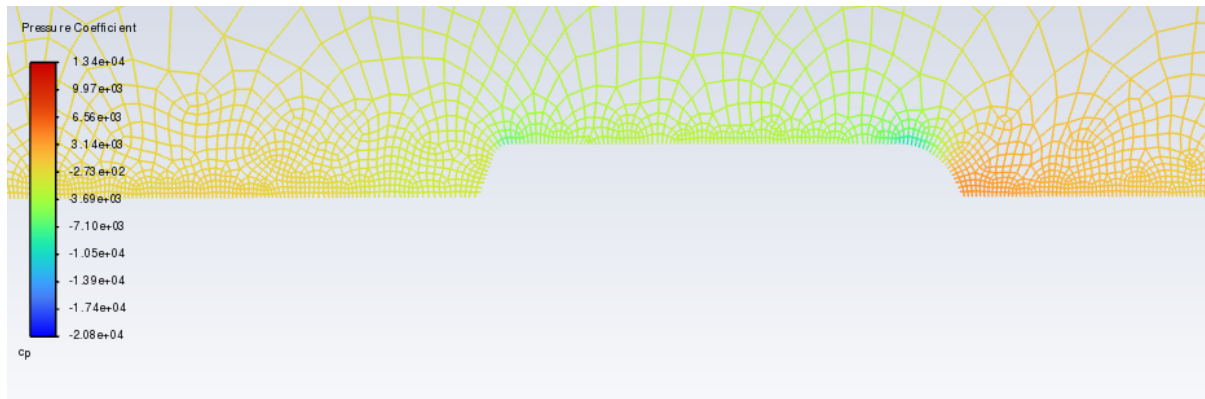


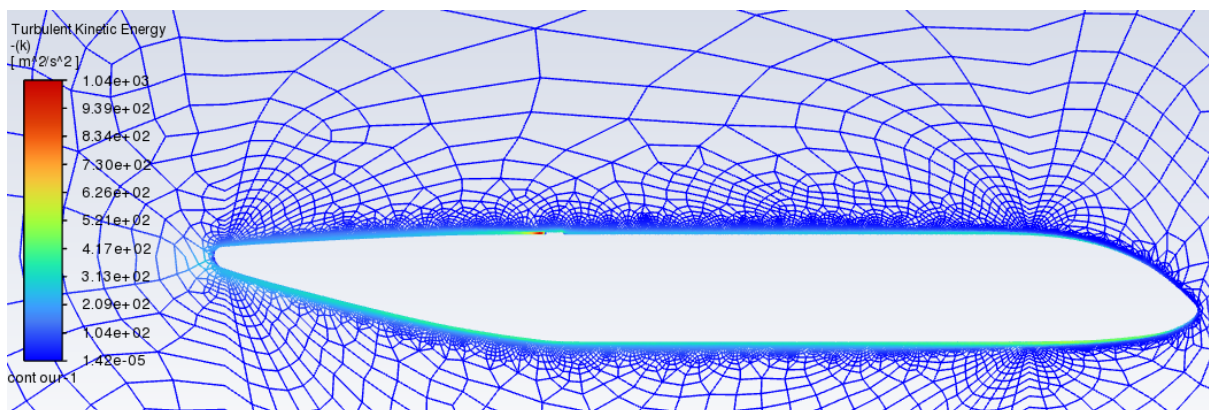
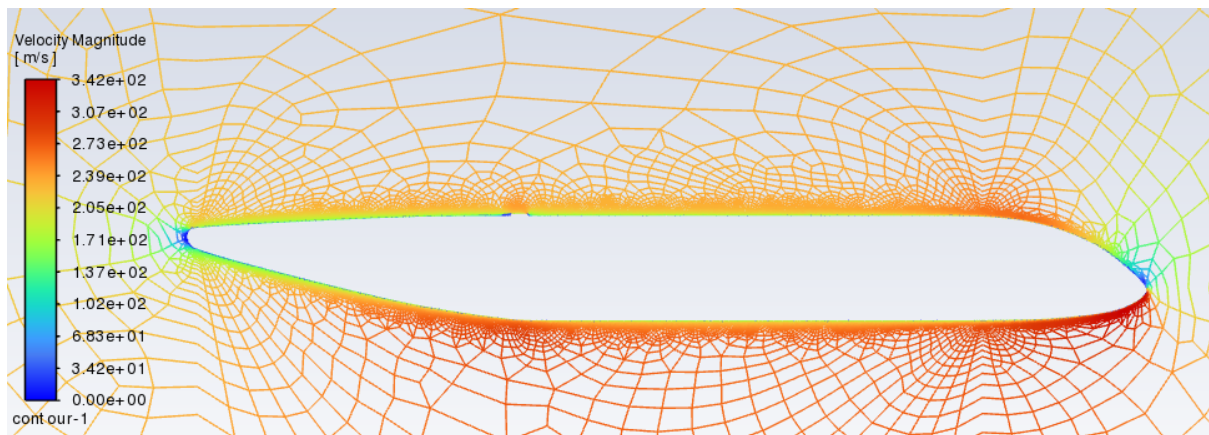
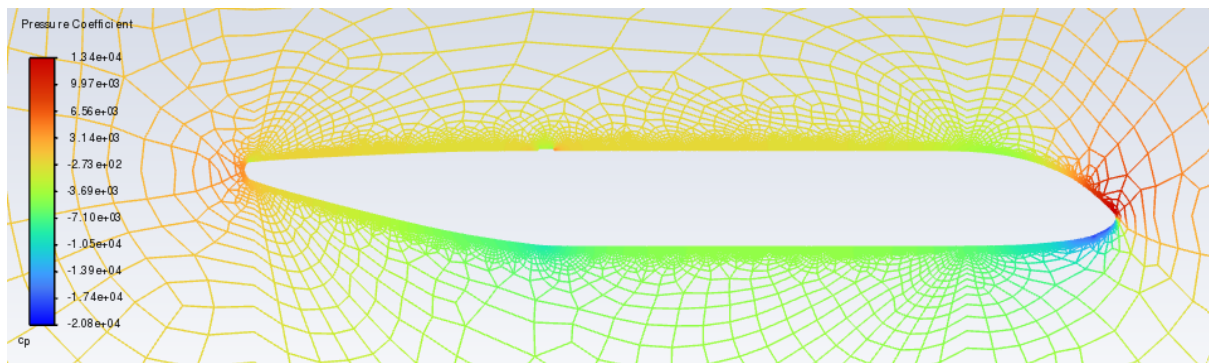
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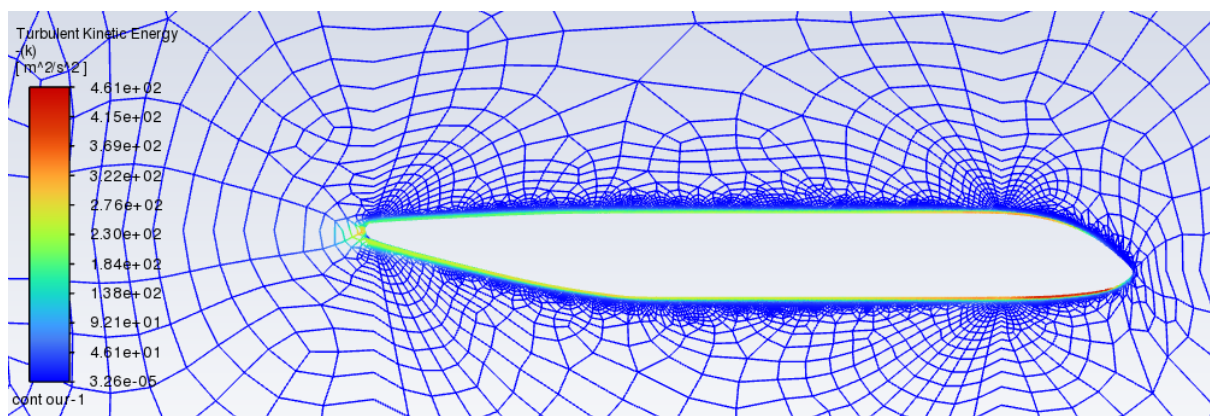
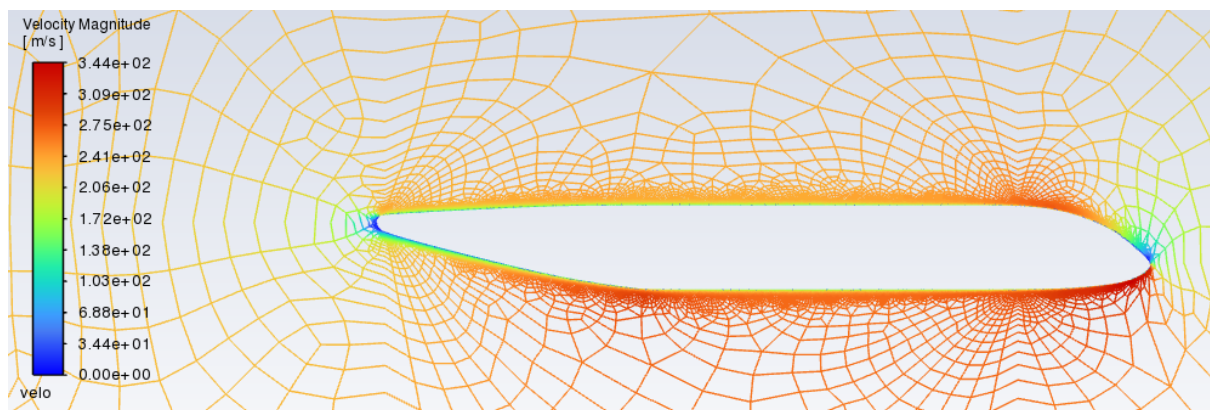
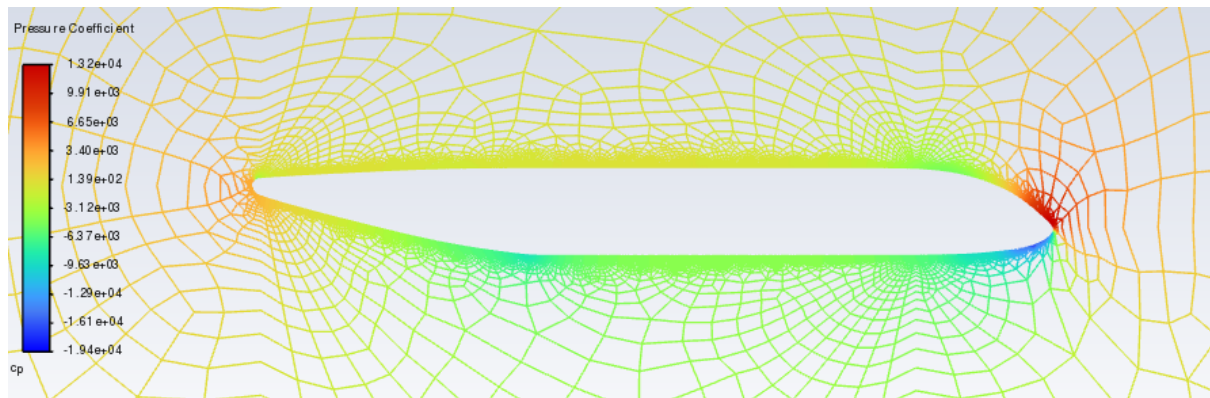
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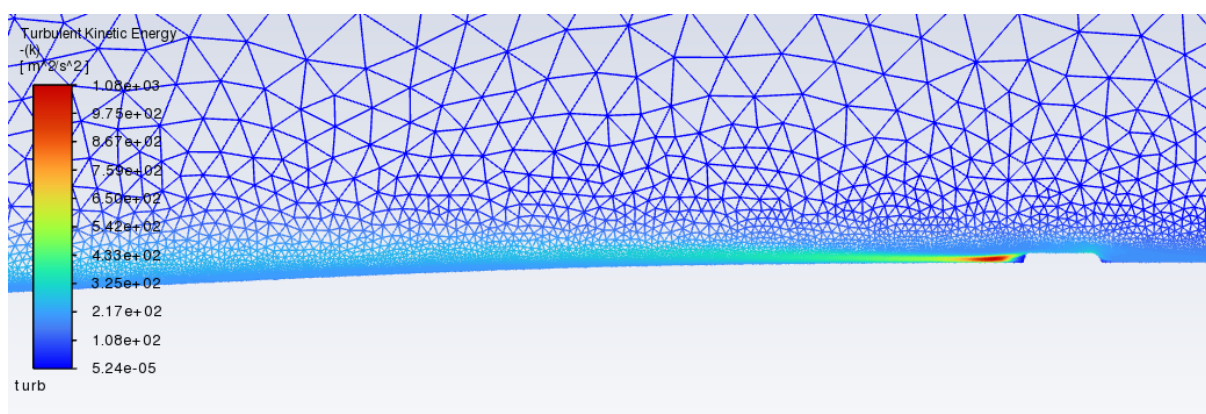
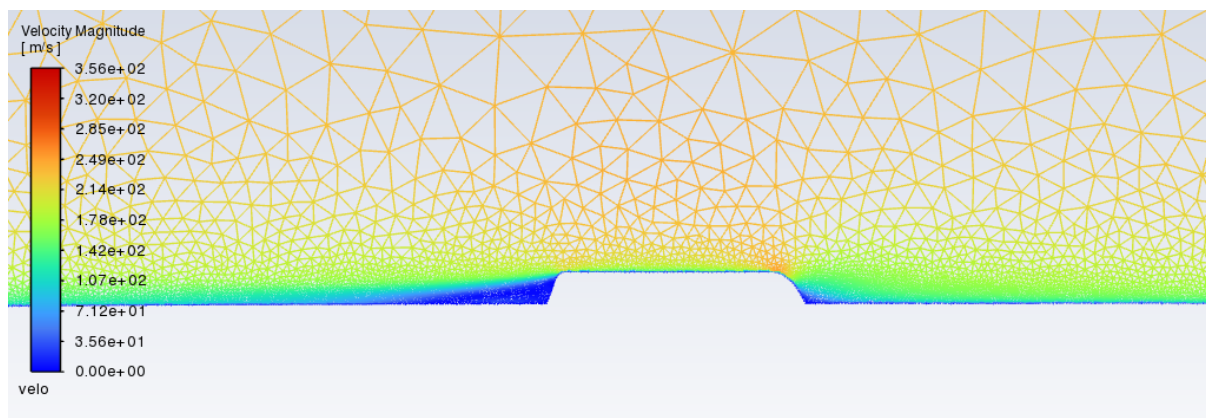
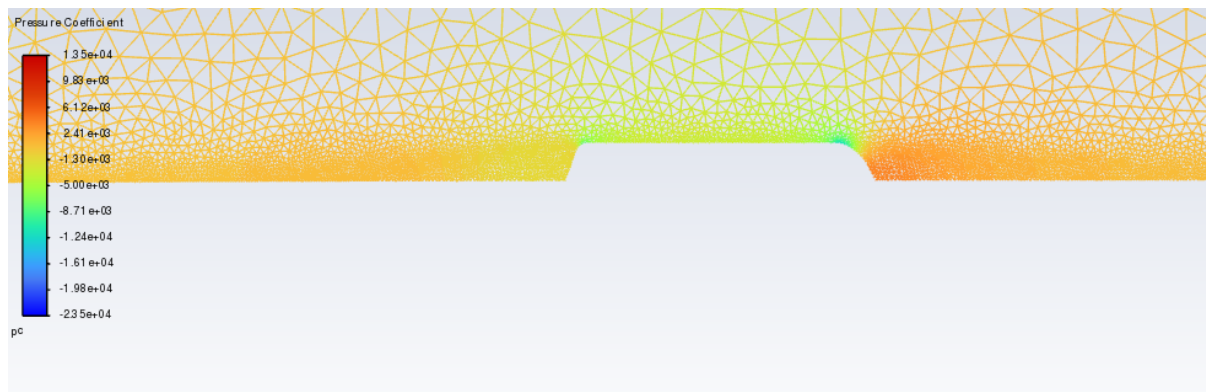


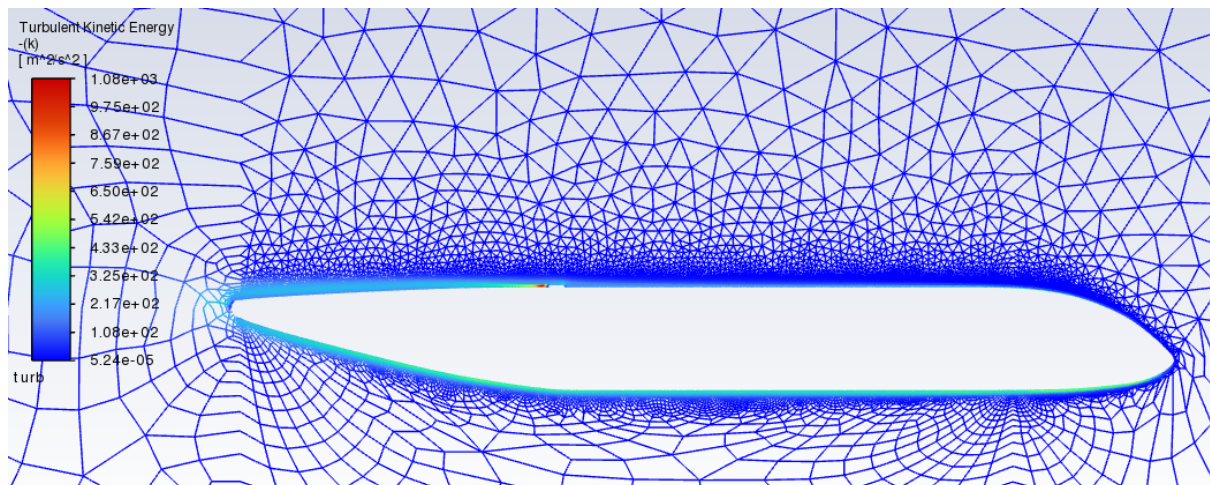
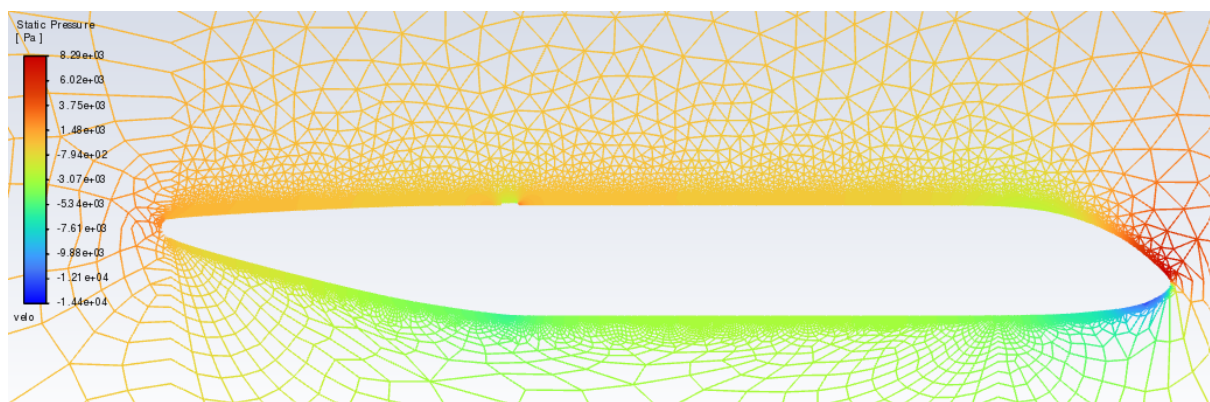
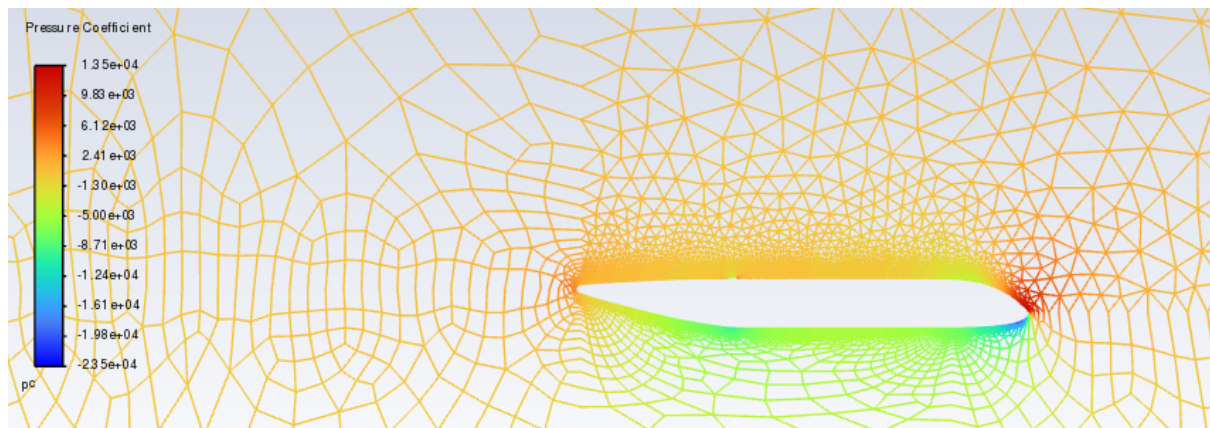
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.6.2 No antenna

